

When We Police

Real-Time Surveillance and Investigation

Our officers are expected to know the unknown and see the unseen. They are making split-second decisions based on limited facts. The more you can provide in terms of intelligence . . . the more safely you can respond to calls.

—Fresno Police Chief Jerry Dyer¹

Beware Your Threat Score

First responders operate largely in the dark. A 911 call for a disturbance or report of violence requires police to approach a home without much information. If you are that responding officer, you might want to know if the person behind the door is a gangbanger or grandmother, a felon or first-grade teacher.

Enter big data and the data brokers who have been collecting billions of bits of personal information about you, your family, and your home. Conveniently, this information is arranged by address and accessible in real time for police responding to your home. The same folks who know you have stable credit, two kids, and a good job and like fine wine and cooking magazines can also predict whether you will be a danger when you open the door. Through big data, police can get the information before they need to act.

Police in Fresno, California, piloted a service called Beware to give them real-time “threat scores” about addresses and people.² Beware searches through proprietary consumer databanks to provide a rough predictive judgment about the 911 caller, the address, or the neighborhood.³ The predictions initially involved color-coded threat levels (red, yellow, or green) and provided some measure of risk assessment for police responding to a scene. Beware’s data comes from commercially available records such as criminal records, warrants, and property

records—in essence, most of the data populating consumer big data systems—and early versions also incorporated social media data, which might pick up on threatening statements, gang associations, or other violent comments.⁴

Not much is known about how Beware determines threat levels. The service is owned by Intrado, a subsidiary of West Corporation, and has kept the threat-score algorithm out of public view.⁵ In community hearings about the technology, the Fresno police chief admitted that Intrado had not even informed the police about how the threat scores were calculated. But, like many things with big data, the secret is not really much of a secret.⁶ Data about arrests, convictions, gun licenses, neighborhoods, crime rates, and the like remain readily available in the consumer data marketplace. In fact, if the same first responder simply typed the 911 caller's address into an existing police database, many of the same pieces of information could be obtained and the same inferences drawn.

Predictions, of course, have consequences. If police officers have information that a violent felon lives in a particular home, the officers are more likely to be on guard. Police are more likely to react to the danger and are more likely to shoot at a perceived threat. Thus, a high threat score might well produce hostile or even deadly results. A threat prediction literally and figuratively colors judgment about the relative danger.

Unsettlingly, the data can be wrong. In the same public hearing about the use of the technology, a Fresno City Council member asked the police chief to run his address through the Beware system. The council member's house came back with an elevated "yellow" threat level.⁷ While no one suspected this particular official of being a danger, a responding police officer would still treat him as such. No one at the hearing could explain the elevated threat level. These concerns and other civil liberty issues caused the Fresno Police Department to back away from continued use of the Beware technology, reducing reliance on color-coded threat levels and social media information.

This chapter explores how new surveillance technologies shift when police act. Immediately available information, all-seeing surveillance, and large-scale tracking capabilities with automated alerts help speed up decision making. For police officers needing to make split-second decisions or investigators needing to sort through stacks of files, big data can offer significant efficiency benefits. But data available in real time

helps only if the information is accurate. As with the Beware system, questions of reliability, transparency, and bias remain and must be addressed. In addition, constitutional concerns arising from video surveillance systems challenge traditional Fourth Amendment understandings of personal privacy in public.

Real-Time Domain Awareness

A car drives into lower Manhattan. A man with a red shirt gets out with a bag. He walks a few blocks and leaves the bag on the street. He hops in a different car and drives away. In any other city, this action would be one of a million anonymous acts, untraceable, lost in the bustle of a big city.

But not in New York City. Not with the Domain Awareness System (DAS)—a partnership between the NYPD and Microsoft. The DAS links approximately 9,000 closed-circuit surveillance cameras for real-time monitoring of lower Manhattan.⁸ The video feeds go directly to a digital alert system that automatically tracks for suspicious behaviors (such as leaving a bag on the street). Cameras pick up images of the car and automated license-plate readers record every car that enters the area. These 500 license-plate systems connect with DMV records, police watchlists, open warrants, and terrorist databases and all of the personal information associated with these databases.⁹ The recorded video can be replayed to track the direction, location, and movements of the suspect, and the technology can even search for descriptions, such as “all people wearing red shirts near the New York Stock Exchange.”¹⁰ Still photos of matching people can be pulled up with one search, tagged to location, time, and date.

Under the DAS, as soon as the man abandons the bag, an automatic alert triggers. Cameras within 500 feet of the location can be reviewed, and officers can play back the path of the man getting out of the car.¹¹ The car itself could be traced back to the point it entered the network of surveillance cameras, and using the automated license-plate tracking systems, information about the owner or whether the car was stolen could be obtained. Similarly, the getaway car could be tracked and analyzed. Camera data going back a month can be studied about the area, so that historical information about whether this was a terrorist or a tourist

could be relayed to responding officers. To augment the data sharing, NYPD invested in 35,000 smartphones so patrol officers could have access to DAS on patrol.¹²

More limited versions of the circle of cameras have been tested in Los Angeles, Fresno, and other cities. Fresno even sought to link police body-worn-camera videos to the network of surveillance.¹³ With such linked camera systems, the normal limitations of human observation and memory become all but irrelevant. All-seeing digital technologies speed up police reaction time and also prolong investigative capabilities.

For first responders, real-time surveillance can provide a wealth of helpful information. Police can know who is on the scene, what has happened, and whether anyone poses a danger. Police regularly must sort out friend and foe, victim and offender, in the middle of highly emotional incidents, so visual assistance from supervisors able to review what actually happened before police arrived can help make better decisions. In addition, other first-responder medical services can be provided before police arrive, shortening ambulance calls or the time to receive emergency services.

Police response time improves through automation. Automated suspicion algorithms using artificial-intelligence capabilities can turn surveillance cameras into digital spies able to recognize suspicious patterns and alert the police to the crime.¹⁴ The same type of pattern-recognition algorithm that can identify an abandoned bag on the street can also identify other suspicious human activities. For example, in East Orange, New Jersey, police adopted Digisensory Technologies' Avista Smart Sensors to automatically monitor the streets.¹⁵ These smart cameras have been trained to look for patterns of suspicious activity, processing 60 billion instructions a second.¹⁶ For example, repeated, frequent hand-to-hand transactions on a corner may signify a drug dealer. A quick movement toward a person and then running away might signify a robbery. Sensors monitor the actions on the street and then alert police if the pattern observed matches a pattern the sensor has been trained to identify. Once the sensor triggers, the nearest two officers are automatically alerted to the potential crime, time, and location. Cameras can trace back the scene, so commanders at headquarters can make sure the algorithm got it right. These automated suspicion algorithms guide police to the scene, providing clues that would be missed by routine police patrols.

Similar algorithms exist to identify gunshots from audio sensors. One technology—ShotSpotter—provides automated reports of gunshots, alerting police to possible violent crime before it is reported by human witnesses.¹⁷ As soon as the highly sensitive microphone hears the sound of a gunshot, police are dispatched to the targeted location. Victims can be found faster, perpetrators caught, and witnesses interviewed. These automatic sensors have been deployed in Washington, D.C., Boston, Oakland, San Francisco, and Minneapolis.¹⁸ Crime is being responded to in real time, usually more quickly than any human could react.

Tracking Cars, Faces, and Spaces

License plates are unique identifiers, linking cars to their owner. Automated license-plate readers (ALPRs) are camera-like devices that can scan a license plate and automatically compare the license to a database of active warrants, stolen cars, unpaid tickets, or any other variable.¹⁹ Affixed on the top of police cars, ALPRs can scan thousands of license plates a day. In practice, a police officer can drive on a highway, pass a car associated with an open warrant or registered as stolen, and immediately receive a real-time alert.²⁰

The digital record of ALPRs has grown exponentially, as police vehicles record millions of license plates each year. Sophisticated enough to read one license plate every second, ALPR systems can generate city-wide maps of car locations. By driving through parking lots, apartment complexes, and ordinary streets, the system automatically memorializes where particular cars were at particular times. An aggregated ALPR map can, thus, link cars and owners by time, date, and location. Over time, the accumulated tracking of cars provides clues about travel patterns, habits, and the actual location of cars at certain times.²¹ If police suspect that a particular house is used for drug dealing, the repeated sighting of a license plate might be a clue to suggest involvement in the drug trade. In emergency rescue situations like an “Amber Alert” child abduction, searches for car license plates can lead to clues about the location of the suspect. For investigations, police can find possible witnesses to a crime. Such a database of locational information, of course, presents a real privacy problem because everybody with a car can be sucked into the data stream. Journalists have requested their own ALPR data and found their

personal car tracked, with photos of their car and family and a revealing pattern of personal activities.²²

What ALPR does for cars, facial-recognition technology does for people walking on the street. Advanced surveillance cameras in Los Angeles have the capabilities to scan and compare live facial images to mug shots in police databases.²³ These fixed cameras can scan faces from 600 feet and can match faces with anyone in the police database.²⁴ Like a car with a license plate, a person wanted for criminal activity can be automatically identified using facial-recognition technologies. In addition, a digital map of past sightings is available for later investigatory use, so that should a crime occur near a camera, police can scroll back the video and identify all the people who walked past the camera at the relevant time.

Facial-recognition technology offers virtual fingerprinting for anyone in front of a camera. The FBI possesses a database of 30 million photographs.²⁵ State databases contain millions more.²⁶ Some photographs come from civilian drivers' license photos, meaning that millions of innocent people are regularly put in virtual criminal photo spreads.²⁷ Many other photographs result from criminal arrests. In just under five years, the FBI requested 215,000 searches of facial-recognition databases, with 36,000 searches of state driver's license databases.²⁸ When connected with real-time video capabilities and sophisticated facial-recognition technologies, a new web of visual surveillance may soon spread over cities.

The growth of police-worn body cameras provides additional identification capabilities. Police-worn body cameras are small cameras usually affixed to the front of a uniform, and some police departments even have adopted software that allows personal smartphones to double as body-worn cameras.²⁹ These cameras create a continuous feed of the people police come in contact with during their daily routines.³⁰ The video can be used to identify who, where, and when police made contact with a particular person. Next-generation body cameras will include real-time facial-recognition technology that will allow police to know about active warrants, prior violence, or generally peaceful conduct (just like the fixed cameras).³¹ While current limitations on battery power and computing requirements have made facial recognition through body cameras more aspirational than real, companies promise that in

the next few years this capability will be operational.³² In addition, the facial-recognition technology itself is growing more sophisticated by the month. Facebook's facial-recognition program DeepFace boasts of a 97% accuracy rate of identifying faces across its platform, with new improvements happening every year.³³

If DAS and body cameras are not invasive enough, aerial cameras provide the ultimate mass-surveillance tool. Flying high overhead, able to record entire neighborhoods for hours at a time, aerial cameras like Persistent Surveillance Systems can watch crime in real time and record the patterns of all vehicles, people, and events below.³⁴ *Washington Post* reporter Craig Timberg vividly described the plane's observational capabilities:

Shooter and victim were just a pair of pixels, dark specks on a gray streetscape. Hair color, bullet wounds, even the weapon were not visible in the series of pictures taken from an airplane flying two miles above.

But what the images revealed—to a degree impossible just a few years ago—was location, mapped over time. Second by second, they showed a gang assembling, blocking off access points, sending the shooter to meet his target and taking flight after the body hit the pavement. When the report reached police, it included a picture of the blue stucco building into which the killer ultimately retreated, at last beyond the view of the powerful camera overhead.³⁵

Add sophisticated audio capture, and these aerial camera systems or drones provide the potential for cities to be surveilled in real time and for days at a time. Similar to the Domain Awareness System, police can roll back the tape and watch how the crime developed, where the suspects fled, and patterns of behavior before and after the suspects got to their supposed safe houses. Persistent Surveillance Systems planes have flown missions over Baltimore, Los Angeles, Indianapolis, Charlotte, and other cities.³⁶ Cars or people can be digitally identified and tracked over time. Time and movement can be reduced to searchable data points. A digitized history allows time's normal passing to be recaptured. And all of the information (tens of thousands of witnessed moments) can be (and has been) turned over to police interested in solving crimes.

Real-Time Investigation

The first 48—homicide investigation lore holds that the first 48 hours of an investigation remain the most important. If you speed up the investigation, you close more cases.

Across the country, police departments are developing real-time operations systems to investigate crimes. These nerve centers centralize information gathering,³⁷ linking video feeds, calls for service, crime maps, and real-time police patrols to allow commanders to respond to ever-shifting crime patterns immediately. An uptick in robberies in the afternoon can lead to additional police patrols at night.

But the game-changing technology involves big data networks of investigative leads. Sarah Brayne conducted a groundbreaking two-and-a-half-year case study of the LAPD's embrace of data-driven technology.³⁸ Brayne describes how larger and larger data sets have made police investigation faster, more accurate, and data centered.³⁹ She describes a homicide investigation that began with a body dumped at a remote location. The victim appeared to be a Los Angeles gang member, but no witnesses could be located. Fortunately, an automated license-plate scanner recorded license plates near the location. By narrowing down the time of day, police focused on a single license plate from Compton, California. By running the tags in a database and cross-checking the owner's name with a gang database, police generated a suspect who happened to be in a rival gang from the person killed. The suspect's vehicle was searched, and incriminating evidence was recovered linking the suspect to the victim.⁴⁰ Case closed.

Networked systems allow police to process more data, much more quickly. Palantir developed an integrated data system that allows LAPD to search for people, automobiles, homes, cellphones, email addresses, locations, friends, associates, family, or employment from among the various data sources already in the possession of the department.⁴¹ If any link can be connected between two of these things, it can be digitally mapped. If investigators want to extend out a second or third link, entire groups can be connected. Gang members, people with open warrants, and others targeted for investigation can all be linked together, so you could see all the home addresses of a particular gang or the travel patterns of three known drug distributors.⁴² For big-picture investigation,

this allows police to see the nature of crime networks in the city (and across multiple jurisdictions).⁴³

Non-law-enforcement data also populates the investigative databases. Information from social services, health services, mental health services, foreclosures, social media, utility bills, and even stored phone records from pizza chains (used to speed pizza delivery) can all be potentially linked up.⁴⁴ Such a comprehensive data-aggregation system might run into privacy issues, and the inclusion of health information would certainly add additional legal complications; but as a technical matter, this data is available and can be integrated into aggregated police systems.

At a minimum, the LAPD database can be used to target particular crimes. If, for example, police wish to track a car suspected of being involved in human trafficking, police can find all the times the automated license-plate reader tagged the car. The pattern can be visualized on a map. This connecting thread might lead to finding trafficked people or routes used for smuggling people between houses. Or if police learn about three particular houses used for prostitution, a “geofence” (electronic sensor that alerts when crossed) can be created around the houses,⁴⁵ such that if any single car shows up at all three houses, an inference can be drawn that the individuals in the car are involved in the suspected trafficking activity.

All of this can be done in real time. Instead of conducting a stake-out, an automatic alert system can be created to ping the officer that a particular car has entered an area or person has come into contact with police. Literally, an email goes out informing the officer that the wanted person has just made contact with a police sensor. In addition, similar to the Beware system, all of this accumulated data about a location can inform officers responding to a 911 call or effectuating an arrest. Data about the area, the people who live at an address, past crimes in the neighborhood, and other data that has been entered into the system can be provided almost instantaneously. Automation pushes out tactical information constantly, quickly, and conveniently to officers in the streets.

Race for Data

Proactively studying crime means mapping it. In the most heavily surveilled areas of Los Angeles, field contacts, ALPR, and facial-recognition

records pinpoint the location of people and cars. As data points, these basic locational facts can be studied over time. As clues to solve crimes, these facts can be useful for identifying witnesses, tracking suspects, and even providing (or debunking) alibis. Further, because the information is not only individualized but associational, link analysis can connect many other people to the groups involved in particular group crimes.

Communities of color bear the brunt of these tactics. By and large, it is poor people from minority backgrounds who are stopped by police. By and large, it is people of color who are populating the growing police databases. If these racially skewed databases of past police contacts become the justification for future police contacts, then biased data collection will distort police suspicion. Worse, when linked to commercially available big data systems, citizens might be inappropriately stereotyped based on overbroad generalizations.

Similarly, the choice of where to place the surveillance systems can be seen as discriminatory. The Stop LAPD Spying Coalition argued that by targeting Skid Row, the facial-recognition technology mapped only poor people of color living and working in the community.⁴⁶ In Baltimore—home to a police department under federal investigation for racial discrimination—the police used data from the aerial Persistent Surveillance Systems over West Baltimore (a predominantly African American community).⁴⁷

The choice of targeting communities of color matters, because race actually distorts the accuracy of certain technologies. For example, facial-recognition matching works by creating models of faces, digitally broken down into tiny pixelated nodes.⁴⁸ With current technology, every face can be digitally deconstructed to 80 nodal points and thus matched with other faces in a database system. Facebook recently developed technology that can identify a face out of the 800 million photographs on its network in under five seconds.⁴⁹ Facial-recognition algorithms are trained by looking at faces, but because algorithms are trained on existing photos, certain racial bias seeps into the process.⁵⁰ As Clare Garvie and Jonathan Frankle wrote in the *Atlantic*,

Algorithms developed in China, Japan, and South Korea recognized East Asian faces far more readily than Caucasians. The reverse was true for algorithms developed in France, Germany, and the United States, which

were significantly better at recognizing Caucasian facial characteristics. This suggests that the conditions in which an algorithm is created—particularly the racial makeup of its development team and test photo databases—can influence the accuracy of its results.⁵¹

Another study of facial-recognition technology found racial disparities between African Americans and Caucasians. The technology failed to identify the correct person twice as often when the photo was of an African American subject.⁵² This inaccuracy can have one of two results. First, because the facial-recognition technology cannot find a match (even though there is a match in the data set), a person who should be identified is not identified (the guilty go free). Second, because the facial-recognition technology cannot find a match (because there is no match in the data set), the algorithm nevertheless suggests the person who is the closest match (an innocent is flagged). This closest match becomes the target of investigation,⁵³ and in the United States, due to the embedded racial disparity of who gets arrested (and thus who is in the data system), this error will fall on racial minorities more than on whites. In any algorithmically dependent project, these errors can have discriminatory impacts about who is stopped or even arrested.

Race does not need to have a distorting effect on policing. In New York City, the Domain Awareness System focuses its attention on downtown New York, in the financial center. Such placement avoids direct allegations of racial (or economic) bias correlating with minority communities or that the cameras were intended to target high-crime areas (assuming Wall Street does not count as a traditional high-crime area). Because of the commercial nature of the area and the terrorist attacks of September 11, 2001, use of the Domain Awareness System in Manhattan has avoided much of the criticism heard in Los Angeles.

Race, policing, and surveillance collide when police use mass-surveillance technologies to monitor domestic political unrest. In both Ferguson, Missouri, and Baltimore, Maryland, the FBI used sophisticated aerial surveillance to monitor protesters challenging police use of force.⁵⁴ These surveillance flights—initially secret—were eventually exposed by curious citizens who saw small planes flying in odd patterns. In Baltimore, federal agents accompanied by local police flew planes

equipped with high-tech cameras to provide live video feeds and even lasers to capture photographs in darkness.⁵⁵ According to the *Baltimore Sun*, the surveillance aircraft made ten flights over the protests during the days following the death of Freddie Gray in 2015,⁵⁶ confirming activists' complaints of ongoing surveillance and tracking by federal law enforcement agencies.⁵⁷ Obviously, use of such technologies against peaceful protests raises difficult constitutional questions about the limits of police surveillance, the importance of free speech, and the need to ensure civil order. These questions are made more difficult because of the racial cast of the surveillance, targeting not just communities of color but members of the Movement for Black Lives.

Unreal Reliance

Big data technologies are changing the pace of police response. Driving these technologies are automated capabilities that can search information faster than human investigators can and that can make that information usable in real time. The timing of when we police, however, has its downsides as speed does not always encourage accuracy.

Denise Green learned the hard way that automated suspicion can fail. One night, Green, a 47-year-old African American woman, was driving her burgundy 1992 Lexus ES 300 on Mission Street in San Francisco.⁵⁸ A police cruiser with ALPR technology passed by and alerted to a stolen car. Green's license-plate number was 5SOW350, but the ALPR misread the license plate as 5SOW750. Due to darkness, the photograph normally used to confirm the automated report was "blurry and illegible."⁵⁹ The officer driving the cruiser radioed ahead regarding the stolen car, and a second officer, without double checking the license plate (or the fact that the stolen tag 5SOW750 corresponded with a gray GMC truck, not a burgundy Lexus sedan) performed a "high-risk stop." This stop entailed Green being forced out of the car at gunpoint. Green, a municipal driver for the city with no convictions, was required to kneel in the middle of the street and be handcuffed by six armed officers.⁶⁰ She sued the police for the violation of her constitutional rights as a result of the mistake. While police protocols did suggest double checking and confirming the license plate before stopping the driver, police just deferred to the automatic alert without a second thought.

“Garbage in, Garbage out. Everyone knows that much about computers; you give them bad data, they give you bad results.”⁶¹ So begins another federal case—written by then-judge Neil M. Gorsuch—dealing with the accuracy of a computerized state license-plate database. In this case, a state trooper in Kansas stopped Antonio Esquivel-Rios because his car’s temporary Colorado tag did not show up in the database that was routinely used to search for licensed cars.⁶² It turns out, however, that temporary tags from Colorado could not show up in the Kansas database because temporary tags were not routinely loaded into the Kansas system.⁶³ The police officer based his stop solely on the database response, but the database was an unreliable source of information. In fact, only at the subsequent court hearing did it come to light about the types of licenses included in the shared database and the fact that such temporary tags were never loaded in. The police officer, however, did not know that fact and relied on the incomplete information in the database, resulting in the erroneous stop. Such database errors are unfortunately common. Over the past few years, the U.S. Supreme Court has decided four cases involving arrests based on faulty arrest warrants in police databases.⁶⁴ Data-driven suspicion is only good as the data in the system.

Accuracy concerns also exist with consumer data repurposed for criminal investigation. The Beware program in Fresno, for instance, relies on similar commercial data compiled by data brokers. While data brokers have proven quite useful in making sure we have the appropriate catalogues in the mail, it does not matter very much that they occasionally get things wrong. But when the Beware system provides inaccurate (or merely out-of-date) information to police responding to an emergency, the errors can turn deadly. If, for example, a 911 call returns information that a violent felon lives in the house, any responding officer would be wise to pay attention. But the heightened vigilance could turn tragic if the data is wrong, is dated, or causes the officer to overreact out of fear in a situation that did not call for it. Maybe the felon moved away. Maybe the address had merely been given to receive mail, and his grandmother lived in the house. Maybe the former felon had turned his life around. The consumer data captures none of these realities and yet directly impacts how police perceive the situation and how they respond.

Similarly, facial-recognition technologies have yet to be adequately tested for accuracy. The Government Accountability Office (GAO) issued a critical report on the FBI's facial-recognition technology policies, castigating the FBI for failing to assess the accuracy of its searches.⁶⁵ The FBI did not assess the "false positive rate," did not conduct operational reviews of the actual use of the technology, and did not assess the accuracy of external face-recognition systems from partners.⁶⁶ Simply stated, the most advanced and sophisticated user of facial-recognition technology could not validate the accuracy of the system. In 2016, the Georgetown Law Center on Technology & Privacy issued a nationwide report titled "The Perpetual Line-Up" detailing the growth of facial-recognition technologies.⁶⁷ The report found that 117 million (one in two) Americans have their images in these law enforcement database systems.⁶⁸ Many driver's license photos now can be used in police photo arrays to match against suspected criminals. These local, state, and national databases of images (of largely innocent people) remain unregulated with few legal (or even regulatory) protections.⁶⁹ Using this system, police conduct automated searches for suspects every single day.

The automated nature of the information creates two interrelated problems. First, the officers relying on the automated tips or matches have no way to double check the black-box data. This is a real "black data" problem, as the systems have not been designed to explain the match or data hit but only to flag the information. Almost by design, police can only defer to the data since there is no way to see beneath the result.

Second, the speed at which the information is supposed to be used reduces the ability to check for accuracy. In a non-big-data world, while it is certainly possible for a police officer to obtain bad information about who lives at a particular address, the usual human process of obtaining that information (asking fellow officers, looking through records, personal observation) might provide more contextual understanding than an automated color-coded threat score. With automated threat scores or automated license-plate scans or automated facial-recognition alerts, police will have a tendency to simply accept the information as true without questioning it or the assumptions behind it. Worse, they will lack the ability to interrogate why or how the data was created.

The Time-Machine Problem

Stephen Henderson examined the technological future and realized we have a “time machine problem.”⁷⁰ Available mass-surveillance technologies, like Persistent Surveillance Systems, now can observe, record, and digitize movement in real time and thus can be used to investigate after the fact. Police can see into the past, piecing together data about cars, people, and movements. Drones, ALPR, the Domain Awareness System, and old-fashioned surveillance cameras all provide wonderfully helpful digital time machines.

Such surveillance capacity presents a real threat to privacy, because in order to collect the time-machine data on a particular suspect, the rest of the population also must be recorded. Associational freedoms can be threatened by the mere possibility of such mass surveillance. As Neil Richards has written, “The gathering of information affects the power dynamic between the watcher and the watched, giving the watcher greater power to influence or direct the subject of surveillance. It might sound trite to say that ‘information is power,’ but the power of personal information lies at the heart of surveillance.”⁷¹ With all-seeing surveillance, police power expands, and citizens’ freedom shrinks.

Big data police surveillance, thus, presents a stark privacy problem. The digitization and ability to search and recall particular data points changes the traditional physical limitations of policing. In doing so, it also distorts the constitutional protections of citizens.

Traditionally, the U.S. Constitution through the Fourth Amendment has not provided much protection for activities that occur in public.⁷² Think about it: if you walk outside your house, why should you expect any privacy? You are in public, and a nosy neighbor, paparazzo, or police officer can observe what you do. So for years, police could set up surveillance cameras, trail you in unmarked cars, and watch with whom you meet without any constitutional concerns. Most modern police investigation takes advantage of this absence of constitutional protection from ordinary observational surveillance in public.

The question becomes, does this analysis change with pervasive surveillance?⁷³ After all, police have the technology to record your every public movement (with drones), listen to every public conversation (with high-tech audio devices), and track your every movement (with

GPS technology). The fact that police can obtain this information as a technical matter does not mean that they should be able to obtain this information from a constitutional point of view. This was—more or less—the issue presented in a recent Supreme Court case, *United States v. Jones*.⁷⁴

Antoine Jones owned nightclubs and other business interests in Washington, D.C. He was a successful man about town but also, according to police, a major figure in a drug distribution ring.⁷⁵ The difficulty for police was being able to connect Jones to the drugs. In an effort to make that connection, police attached a GPS device to Jones's Jeep Cherokee and monitored it for 28 days. As the Court explained, "By means of signals from multiple satellites, the device established the vehicle's location within 50 to 100 feet, and communicated that location by cellular phone to a Government computer. It relayed more than 2,000 pages of data over the 4-week period."⁷⁶ Fortunately for police and prosecutors, the GPS coordinates of the Jeep linked Jones to one of the stash houses suspected of containing the illegal narcotics.

During the criminal prosecution against Jones, he challenged the use of the GPS device placed on his Jeep. He argued that this type of tracking without a valid warrant constituted an unreasonable search and, thus, violated his Fourth Amendment rights. The government responded that all of Jones's movements were in public, so Jones really should not have had any expectation of privacy in his travels. After all, police could have followed him using old-fashioned surveillance techniques on public roadways without any constitutional concern.

Jones presented the Supreme Court with a choice between the traditional understanding that people do not have much expectation of privacy in public and the potentially invasive 24/7 surveillance capabilities of new tracking technologies. If the Court sided with the government, it would mean that police could track all cars at all times without any judicial approval. It could mean that big data monitoring in public would have no constitutional limit.

The Court unanimously rejected the government's argument, although it divided on its reasoning. The majority of the Court, led by Antonin Scalia, held that placing the GPS device on the car was a search because it physically invaded Antoine Jones's property.⁷⁷ By physically attaching the GPS device to the Jeep, police trespassed on Jones's private

property. Other justices took a more technologically sophisticated view of the problem. These justices reasoned that the long-term nature of the tracking violated the Fourth Amendment because it violated Jones's reasonable expectation of privacy. These justices, led by Sonia Sotomayor, offered the first real analysis of the danger of the time-machine problem.⁷⁸

The privacy fear, as articulated by several justices, stemmed from the long-term aggregated nature of the data collection and use. The justices reasoned that long-term surveillance made almost effortless by new technology presented a different privacy threat than old-fashioned surveillance in public did. As Justice Sotomayor summarized,

GPS monitoring generates a precise, comprehensive record of a person's public movements that reflects a wealth of detail about her familial, political, professional, religious, and sexual associations. . . . The Government can store such records and efficiently mine them for information years into the future. . . . And because GPS monitoring is cheap in comparison to conventional surveillance techniques and, by design, proceeds surreptitiously, it evades the ordinary checks that constrain abusive law enforcement practices: "limited police resources and community hostility."

Awareness that the Government may be watching chills associational and expressive freedoms. And the Government's unrestrained power to assemble data that reveal private aspects of identity is susceptible to abuse. The net result is that GPS monitoring—by making available at a relatively low cost such a substantial quantum of intimate information about any person whom the Government, in its unfettered discretion, chooses to track—may alter the relationship between citizen and government in a way that is inimical to democratic society.⁷⁹

Justice Samuel Alito also echoed this concern that big data technologies required a rethinking of Fourth Amendment protections:

In the pre-computer age, the greatest protections of privacy were neither constitutional nor statutory, but practical. Traditional surveillance for any extended period of time was difficult and costly and therefore rarely undertaken. The surveillance at issue in this case—constant monitoring of the location of a vehicle for four weeks—would have required a large

team of agents, multiple vehicles, and perhaps aerial assistance. Only an investigation of unusual importance could have justified such an expenditure of law enforcement resources.⁸⁰

A majority of the Supreme Court, thus, agreed that the type of digital time machine at issue presented a threat to Fourth Amendment principles. Yet, despite the awareness of the dangers of mass surveillance, none of the justices offered a clear answer about the future. Five of the justices were in agreement that long-term (28-day) GPS tracking for a drug crime violated the Fourth Amendment because it violated a reasonable expectation of privacy, but no further specifics about other technologies, other time frames, or other crimes were offered. The Supreme Court answered the question before it and nothing more, leaving for another day how it might address the growth of big data surveillance technologies. We do not know if surveillance for 14 days or four days might change the Court's analysis or if a different crime (murder, terrorism) might affect the outcome. We know that technology has changed the Fourth Amendment but just not how much.

The big data surveillance capabilities detailed in this book vastly exceeds the rather-simple GPS location-tracking device used in the *Jones* case. Fourth Amendment questions will arise from use of ALPR devices and facial-recognition cameras. Fourth Amendment concerns will impact DAS and drone surveillance. But as of now, these constitutional issues remain unresolved, and Congress has largely failed to act.

Future threats of mass surveillance are only growing. *Wired* magazine reported that technology now exists that potentially could link 30 million private CCTV surveillance cameras together in a truly massive surveillance system.⁸¹ Private towing companies have begun equipping their trucks with ALPR scanners to assist with repossession claims and have been willing to share that data with law enforcement.⁸² Cities have seen the capabilities of the Domain Awareness System and are looking to purchase it. In combination, this surveillance network provides a real-time map of our daily activities without any guide to future privacy protections or constitutional limitations. This is the Fourth Amendment question for the future, one that threatens to rewrite existing constitutional understandings of liberty, autonomy, and freedom from government monitoring.

How Big Data Impacts When We Police

Changing “when we police” is not just about real-time investigative capacities but also signals a commitment to proactive information gathering. Constant collection of data to understand community networks of criminal activity has altered what police do on a daily basis.

In doing fieldwork with the LAPD, Sarah Brayne observed the effect that new pre-crime data collection had on policing. Central to the LAPD’s data-collection effort were “field interview cards.”⁸³ Police on patrol would fill out these cards, which included personal information such as name, address, sex, height, weight, “descent” (presumably ethnicity), date of birth, aliases, probation/parole status, phone numbers, and Social Security numbers.⁸⁴ Additional information about “personal oddities,” union affiliation, or gang or club membership can be added on the cards. Other information about the subject’s vehicle (including damage, custom accessories, or stickers) and persons found with the suspect can be memorialized. Importantly, the location and time of the contact (geocoordinates) are recorded, so that all of this information could be saved for later investigative use and uploaded to the system to link suspects together. So, for example, if two people were observed on two separate days in the same vehicle, that link—once reduced to digital form—could be observed in the data. Or if several field interview cards identified individuals from the same gang, the location of these gang members could be digitally mapped over time.

In some cases, the field interview cards were bundled into formal “Chronic Violent Crime Offender Bulletins,” which became one-page dossiers about high-risk individuals.⁸⁵ These bulletins would include a photograph of the subject with identifying information but also would list the history of prior contacts between police and the individual (and his associates). These bulletins were not intended to replace arrests warrants and had no independent legal significance but did end up having a real tactical impact. In practice, the Chronic Violent Crime Offender Bulletins were operationalized into Project LASER (Los Angeles Strategic Extraction and Restoration),⁸⁶ the goal of which was “to target with laser-like precision the violent repeat offenders and gang members who commit crimes in the specific target areas. The program is analogous to laser surgery, where a trained medical doctor uses modern technology

to remove tumors or improve eyesight.”⁸⁷ Like priority targets in New York City, these men became the focus of police attention before any report of a completed crime.

This move to collect data through field interview cards began to have an outsized role in policing. Each police contact ratcheted up the likelihood of future suspicion, because each contact counted toward an informal threat score. So just by being the recipient of police attention, one’s threat score increased. Or, in other words, police could increase suspicion just by being suspicious. Second, the push for contacts encouraged officers to initiate physical conversations and stops with those who were targeted. Even if consensual, these stops felt like seizures for those people being stopped. And information that a person was a “Chronic Violent Crime Offender” likely would tip the balance into finding some reasonable suspicion for a stop. After all, if you are a police officer on patrol and you receive a “Chronic Violent Crime Offender Bulletin” with a photograph of someone who lives on your patrol route, you are going to be suspicious of that person.

This pre-crime information also impacts how police treat citizens. Police approaching someone considered a “Chronic Violent Crime Offender” might reasonably use significant caution and perhaps force, including physical force, to ensure their safety. From the perspective of the suspect, such treatment will be considered demeaning, frightening, and unnecessary. While the suspect may well have gang affiliations and a prior record and may be under court supervision, it might be the case that at that particular time, the suspect is trying to go home, get a job, or care for his family.

To be clear, these police contacts do not involve investigating ongoing crimes or making arrests of completed crimes (there is neither reasonable suspicion nor probable cause). They involve pre-crime data collection, transforming police officers into data collectors for a larger system of information control. Police are policing before the crime and building the investigative network to solve future crimes. While no doubt these regular police contacts also impose a measure of social control over a population likely to be involved in crime, the focus is really about proactively developing data, rather than responding to reported crimes.

A real-time data-collection focus makes the job of a police officer far more complex. In addition to doing all of the normal day-to-day tasks,

police officers must also become data-entry experts, interpreters, and technicians. Data in a data-driven policing system comes from the officers collecting it. As can be seen in the push for data from field interview cards with the LAPD, this means a lot of data entry. While certain start-up companies like Mark43 have offered new digital software to speed up data entry and automate police reforms,⁸⁸ for most officers, the paperwork system is not too different from their father's police force.

Besides the time-consuming nature of the data-entry process, this data focus requires police to see things as the data fields consider things. The cards reduce individuals to categories of race, gang affiliation, gender, and the like, rather than thinking about the individuals as individuals. Suspects become data points as much as people. The focus becomes one of tagging individuals rather than seeing the environments in which they operate. By design, the LAPD system is trying to mark people in time and place, so police become hunters, geotagging suspects for the larger data net. Every patrol becomes a data-gathering mission, rather than a crime-suppression mission.

With more information, police officers must become fluent data interpreters. A constant flow of new information is available to police. Before knocking on a door or stopping a car or talking to a person on the street, a wealth of data must be interpreted. Much of this instantaneously available information is imperfectly sourced, contradictory, and overwhelming. Imagine: a police officer stops a car owned by a convicted violent felon, but one who appears to be employed and lives in a big home in a nice neighborhood. Is he dangerous? How does the additional information help the officer sort through the risks? How can the officer check the reliability of the information? Adding a Beware "threat score" filled with miscellaneous consumer data (or the equivalent) only complicates the decision.

Of course, police have always had to make difficult decisions, but the inputs of traditional policing were much narrower. When Officer McFadden watched John Terry, he saw what he saw with his own two eyes.⁸⁹ His prior experience could inform his personal observation, but much of his decision making came from inside his head and gut. But in a big data world, McFadden would be burdened with all sorts of externally derived facts. He might know the type of neighborhood, down to last week's robbery stats. He might know about John Terry, receiving

not just the available law enforcement information but also what the data brokers have on him. He might have information through cameras about where Terry was beforehand. All of these additional inputs probably assist in assessing whether reasonable suspicion exists for a stop, and most are probably quite helpful; but all are new burdens for officers forced to evaluate the reliability and credibility of the information.

Police officers must become sophisticated data consumers. All the data in the world does not help anyone if line officers do not access the information. NYPD's handheld devices become expensive phones if officers do not master the skills of querying the data. These soft computer skills might also have to go beyond merely using police data to the world of social media or data mining. This reality involves not just technological competence and familiarity but the ability to articulate why officers did what they did and why the data can be trusted. Police officers on the witness stand at a suppression hearing will not do well if their answer to "why did you stop the suspect?" is merely "because the computer told me to." Instead, officers will need to be able to articulate how the data informed their professional judgment and why it makes sense to rely on this type of data for this type of case. Only then will courts accept the value of big data policing to determine the fate of criminal defendants.

The growth of pervasive surveillance capabilities, big data analysis technologies, and aggregated personal information threatens to radically expand when (and how) police investigate cases. A single drone with sophisticated video, audio, and tracking capabilities could change crime patterns and privacy protections in one figurative sweep. While the technology does not quite exist yet, one would be hard-pressed to think of a crime that occurs in public that could not be observed and then investigated with an all-seeing drone. Like the ground-level DAS system, police could digitally search for all hand-to-hand transactions and then observe which ones involved drug dealing. Police could (again with not-yet-operational video capacity) identify the individuals and identify where they came from (their home) and their daily patterns. Big cases like bank robberies or murders could be watched, with all of the players tracked back not just to their home but for the past week. Such a time machine might deter a significant amount of crime. With digital mass collection, the time for police to use this information changes, allowing

quicker responses to emergencies but also longer-term investigations of criminal patterns on the streets.

But the cost of such surveillance is our privacy, because that same technology also monitors political protesters, religious dissenters, and everyone else. This is the dilemma of the “collect it all” mind-set. To catch the bank robber, you also need to capture the photo of everyone who uses the bank. Constitutional and statutory law has yet to navigate big data technologies’ privacy-eroding potential, but the two are on a collision course with an uncertain future.

How We Police

Data Mining Digital Haystacks

Our work is very challenging. We are looking for needles in a nationwide haystack, but we are also called up to figure out which pieces of hay might someday become needles.

—FBI Director James Comey¹

Cellular Circuits

The High Country Bandits robbed 16 small-town banks over the course of two years.² Two men dressed in ski masks, jackets, and gloves pulled off one armed robbery after another across rural Arizona, Colorado, and New Mexico. In a posted “Wanted Bulletin,” the FBI described the bandits’ modus operandi: “The unknown male identified as suspect number one often enters the banks in rural locations near closing time and brandishes a black semi-automatic handgun. Suspect number one then demands all the money from the teller drawers. He obtains an undisclosed amount of money, puts it in a bag, orders everyone on the ground, then exits the banks with a second suspect.”³ The bank robbers netted small amounts of cash and disappeared. Over and over, they managed to stay one step ahead of law enforcement, tacking back and forth through small towns in the West. They forgot just one thing. They were being tracked by their data.

The FBI caught the bandits with a relatively simple digital search.⁴ First, the FBI obtained a court order for all of the cellphone-tower records from four of the most rural victimized banks. Since all cellphones regularly check in with the nearest cell tower, a continuous log of nearby cellphone numbers can be obtained. FBI agents figured that if any single phone number showed up at each of the robbed banks, they would have their suspect.⁵ And they were right. One Verizon cellphone number popped up (out of the 150,000 numbers collected) near three of the four

banks (Verizon did not have a tower near the fourth bank).⁶ Another cell number popped up in two of the four bank locations. And that second number had called the first number on repeated occasions.⁷ The FBI went back to the judge and obtained another court order seeking locational data from the two cellphones in question. By examining the cell-tower records, the FBI could demonstrate that one or both of the cellphones had been near most of the 16 bank robberies. The FBI then tracked the phones to their subscribers, Joel Glore and Ronald Capito, and ended the criminal careers of the High Country Bandits.⁸

A more sophisticated cellphone investigation broke open the largest cash theft in Swedish history.⁹ In a crime right out of the movies, a well-financed ring of international thieves used a stolen helicopter, fake bombs, and armed robbers rappelling down ropes into a locked vault to walk off with 6.5 million in Swedish kronor. As described by Evan Ratliff in his spellbinding article *Lifted* in the *Atavist* magazine, the robbers targeted a cash depot holding the equivalent of \$150 million.¹⁰ The plan involved a coordinated assault via helicopter, high-power explosives, and a violent attack that overwhelmed the guards at the depot. The plan worked flawlessly, except that despite the gloves, masks, stolen getaway vehicles, and other precautions, the criminals could not escape their data trails.

Unlike the High Country Bandits, the international thieves had the good sense not to bring along their personal cellphones to the heist. Instead, they used disposable prepaid phones (“burner phones”) to hide their contacts.¹¹ But even disposable phones leave data trails. One of the Swedish police investigators, Jonas “Jocke” Hildeby, decided to find the needles in the haystack of phone calls. He collected a list of all of the phone numbers processed by cell towers near the cash depot at the time of the robbery. The list included 300,000 calls and 18,000 telephone numbers. Hildeby’s goal was to find a closed loop of disposable phone numbers that communicated only with each other and only around the time of the heist. His theory, which turned out to be correct, was that the robbers would be given burner phones to communicate, only use them to plan the robbery, and then ditch them once the robbery was over. In combing through the data, Hildeby found one such closed circuit of 14 phones. These phones spoke only with each other, did so only in the weeks leading up to the robbery, and stopped immediately after the

robbery. In triangulating the location of those 14 phones, investigators could actually map out the path of the different participants during the course of the robbery.¹² For example, police knew that the helicopter had been stolen from a particular hanger and could track one of the burner phones to that location. Essentially, a digital map could be reconstructed with cellular locational data that mirrored each step of the robbery.

The next step required connecting the burner phones used in the robbery to actual people. The police had identified two potential suspects but needed to connect those men to the phones. Again, data trails revealed the link. Police traced the suspects' personal cellphones using the same tracking technology. Lo and behold, the suspects' personal cellphones were located at the exact same place as the burner phones (before the robbery). The suspects had used personal phones to make personal calls and burner phones to further the conspiracy, but the data trails put both sets of phones at the same place at the same time. At trial, the suspects tried to explain away the coincidence, but the data trails undermined their efforts. Using these cell-tower links, as well as other surveillance, digital, and biological evidence, prosecutors were able to convict the men of the crime.¹³

Big data policing takes advantage of these cellular, digital, and biological data trails by creating new ways to search for connections. This chapter explores how data can be mined to aid criminal investigations. But it also addresses the problems with data error, the racial bias infecting data-driven systems, and the temptation of pure probabilistic suspicion.

Cellular Haystacks

In the United States, police have used cell-tower searches in thousands of criminal cases.¹⁴ In some instances, police simply subpoenaed the phone numbers from cellphone companies, and in other cases, the police directly intercepted the cellphone numbers using "cell site simulators."¹⁵ Cell site simulation technology, formally known as an International Mobile Subscriber Identity catcher (IMSI-catcher) but informally known as a "stingray device" or "D[i]RT-box," mimics a cell tower and tricks cellphones into thinking it is an official tower.¹⁶ Once

tricked, the cellphone reveals its unique identifier, and the cell site simulator captures the location, time, date, and duration of the call.¹⁷ The device is precise enough to find a phone within a few feet.¹⁸

The difficulty is that in order to capture a targeted cell number, the stingray device also has to capture all the other phone numbers in the area.¹⁹ To find the guilty needle in the haystack, you have to collect a lot of innocent hay. And because cell site simulators collect data from all phones in the area, they create massive cellular nets of personal information.²⁰ These simulators can be fixed but also have been mounted on planes, which allows the capturing of cell data from thousands of phones. For example, the *Wall Street Journal* reported that U.S. marshals used small Cessna planes outfitted with DRT-box surveillance tools to find wanted suspects down below.²¹

Sifting through cellphone data can result in remarkably effective policing. On May 9, 2009, an employee at the Central Booking Station at the Baltimore Police Department discovered that her car window had been smashed and her cellphone stolen.²² In a city plagued by serious crime, and without witnesses or clues, these petty thefts generally remain unsolved. Two days later, the thief had been caught. The secret? Baltimore police used a stingray device to locate the stolen phone. But there was a problem. The secret was a secret.²³ The Baltimore police had signed a nondisclosure agreement with the FBI at the behest of Harris International (the maker of the stingray device), which prohibited the police from mentioning the device's use in any public manner.²⁴ So investigators did not mention how they discovered the phone in the police reports. Prosecutors could not reveal the information to the defense lawyers or the judge. This secret surveillance caused problems for criminal prosecutions, and in a few cases, judges threw out the charges because the nondisclosure agreement restricted police officers from testifying truthfully about using the devices. In other cases, prosecutors dropped or reduced the charges.²⁵ In the Baltimore cellphone case, prosecutors dismissed the charges after a few months.

The scope of this cellular sleuthing is growing with the technology. In litigation over the use of IMSI-catchers, a Baltimore police detective testified that the police had used the technology 4,300 times since 2007.²⁶ Some of those cases involved serious charges such as homicide, shooting, or rapes, but others fell on the petty-theft side of the line.²⁷

In Florida, court records show that police have used the devices 1,800 times.²⁸ In Tacoma, Washington, police used it 170 times.²⁹ The federal government currently owns over 400 cell-site simulation devices and spent almost \$100 million on the technology between 2010 and 2014.³⁰ Other jurisdictions also own the technology but have maintained secrecy about the scope of its use.

Metadata

When you know a phone has been stolen, it makes sense to search for the phone's location (and thus the thief). But what if you do not know whether there was a crime, but you suspect that watching patterns of communications could reveal a crime? The capturing of the entire haystack of phone records can be wonderfully revealing for police investigators, because the metadata connecting those numbers reveals otherwise-unseen connections. Metadata is "data about data," which in the phone context means the number, contacts, time, date, and location of the call (but not the content).³¹ Metadata is a big data creation, only useful because powerful computers can crunch the 3 billion phone calls made to or from the United States *each day*.³² Metadata is also created by internet and computer usage, social media posts, digital photographs, and pretty much every digital activity done today.³³

The popular discussion of telephone metadata begins with Edward Snowden's revelations that the National Security Agency (NSA) had been collecting phone-record metadata.³⁴ Although the NSA is focused on the collection of foreign—not domestic—intelligence, through the agency's telephony metadata program, the NSA collected locational data about calls, the IMSI identity of the callers, and the duration of the calls but not the content of the calls.³⁵ NSA would query the data for information about specific, targeted phone numbers. Connections between targeted numbers could be linked in "hops," meaning linking all of the numbers connected to the target phone number. So, for example, an international terrorist target's phone number could be linked to all the people he called, as well as all the people they called (one hop), and then linked to all the people the secondary group called (two hops). The network map "hops" outward, expanding to reveal other possible connections between suspected intelligence threats.

In the litigation over the constitutionality of the NSA program, Edward Felten, director of the Center for Information Technology Policy at Princeton University, detailed the revealing nature of metadata.³⁶ In a legal declaration filed in federal court in the Southern District of New York, he wrote,

Although this metadata might, on first impression, seem to be little more than “information concerning the numbers dialed,” analysis of telephony metadata often reveals information that could traditionally only be obtained by examining the contents of communications. That is, metadata is often a proxy for content. In the simplest example, certain telephone numbers are used for a single purpose, such that any contact reveals basic and often sensitive information about the caller. Examples include support hotlines for victims of domestic violence and rape, including a specific hotline for rape victims in the armed services. Similarly, numerous hotlines exist for people considering suicide, including specific services for first responders, veterans, and gay and lesbian teenagers. Hotlines exist for sufferers of various forms of addiction, such as alcohol, drugs, and gambling.³⁷

Felten also detailed how calls to certain numbers can reveal charitable and political donations, connections to activist organizations, employment, and friendships.³⁸ Sometimes a series of calls can reveal intimate or embarrassing details. “Consider the following hypothetical example: A young woman calls her gynecologist; then immediately calls her mother; then a man who, during the past few months, she had repeatedly spoken to on the telephone after 11pm; followed by a call to a family planning center that also offers abortions. A likely storyline emerges that would not be as evident by examining the record of a single telephone call.”³⁹

A Stanford University study confirmed the invasive power of telephone metadata.⁴⁰ The study involved 823 volunteer participants who provided access to metadata information on 251,788 calls and 1,234,231 text messages.⁴¹ The study sought to create a data set that could test just how revealing a NSA-like metadata program could be. The study provided four main takeaways relevant to big data policing. First, even a “two hop” constraint resulted in linking almost all of the quarter million telephone numbers. Certain “hub” numbers (for example, telemarketers)

contacted so many people that almost all other people could be linked through those shared connections. Second, even though the phone numbers did not have names attached, through the use of publicly available social media and internet searches, the deidentified numbers could be reidentified to find real people. The researchers concluded, “Telephone numbers are trivially identifiable.” Third, by studying the patterns of calls, researchers could determine revealing insights about people’s lives. For example, by studying the pattern and time of phone calls, researchers could determine who was in a romantic relationship. Finally, and most interestingly, the metadata revealed sensitive personal information. The study describes being able to diagnose illnesses, medical conditions, and an interest in firearms just from the numbers called.⁴² Researchers could even develop suspicion hinting at particular crimes, such as calls to a hydroponics store, a head shop, a locksmith, and a hardware store all in under three weeks, which could lead to the inference that the caller had an interest in growing marijuana.⁴³

Despite such privacy concerns, the government has consistently argued that the metadata program—because it was a non-content-based collection system—did not reveal constitutionally protected personal information.⁴⁴ Further, it has argued that because of restrictions on access to and dissemination of the data, as well as the program’s focus on identifying unknown terrorist operations and preventing terrorist attacks, the program did not violate the constitutional rights of United States citizens. Both points are legally important. If metadata does not reveal private data, then it would be hard to claim that the Fourth Amendment’s protection of the reasonable expectation of privacy should apply. Not only did the phone user knowingly share the information with a third-party provider (the phone company), but the collected information does not target personal content (all that is revealed is the data around the phone numbers).⁴⁵ Second, the government argued that a vast amount of data collection was reasonable based on the limited way it was used and the significance of the government’s interest.⁴⁶ While Congress has since banned the bulk collection of telephone metadata under NSA’s previous program, domestic law enforcement can obtain access to metadata for specific cases.⁴⁷

But, really, one does not need the NSA to collect the information. NSA collected the raw information from telephone companies such

as AT&T, which maintained telephone metadata for years.⁴⁸ Literally, trillions of records of phone calls exist in searchable form. Certain law enforcement agencies like the DEA have access to this super database of metadata through a still-secret program called the Hemisphere Project.⁴⁹ The Drug Enforcement Agency has even allowed AT&T employees to work beside DEA agents to more efficiently search the database.⁵⁰ The Hemisphere Project database has been called “a super search engine” or “Google on Steroids” that allows investigators to search connections between numbers and locations on any call that passes through AT&T switches.⁵¹ Without a judicial warrant, law enforcement can obtain targeted information using an administrative subpoena or equivalent judicial order. This metadata connects the dots of social networks and locations of people suspected of criminal activities.

Social Media Mining

A shooting outside a club, a mass protest, a citywide celebration—these events generate instant social media commentary, photos, and video. Individuals at the events can post public comments or images from the events. What if police could zero in on the geographic location and view all of the social media commentary happening in real time, identifying both the people present and the content of the communication? Police in Austin, Oakland, San Diego, San Jose, Santa Clara, and Philadelphia have experimented with Geofeedia,⁵² a software program that allows users to tag, search, and sort social media content for investigation. Other companies such as BlueJay or SnapTrends offer similar social media scraping services.⁵³

The possibilities for investigating social media are alluring to police. Imagine that moments after a club shooting, police use a Geofeedia-like tool to isolate the geographic area around the club and search all of the Twitter, Facebook, Instagram, Picasa, and Flickr posts in real time.⁵⁴ These public comments could be searched for key words (“gun,” “suspects”), and the people in the area could be geotagged, thus memorializing witness locations in real time. Communications between individuals could be linked so that investigators could see who was associated with whom at the scene, and videos and photos of the crime scene could be saved for future use at trial. Police could create a searchable database

of all posts within that geographic boundary, and they could track all future posts of suspects initially identified as being at the scene.⁵⁵

Social media surveillance can also help police investigating high-crime areas. For example, if police suspect a particular housing project as being the home of a drug gang, police can conduct long-term monitoring, intercepting all the social media posts coming from that geographic area. Police can identify all the people posting from the area (both those living there and visiting friends), and police can link those posts to others contacted through social media (developing a web of relationships). The result is a social media surveillance program that can monitor the public content to see if activities, threats, or social commentary suggest criminal activity.

Police in New York and Chicago have long monitored social media, including watching YouTube videos to determine the composition of gangs and related gang activity.⁵⁶ As most gangs involve youth who are fully immersed in social media culture, it is not surprising that gang life bleeds into the digital space. Gangs have used social media to threaten rivals and to stake out turf.⁵⁷ Gangs boast and insult others online, and sometimes threats in videos lead to real-world consequences. Ben Austen in *Wired* magazine interviewed a few of the digitally oriented detectives involved in policing the social media beat, examining the almost “entrepreneurial” way the gang unit in Chicago approaches social media threats.⁵⁸ “The police in Chicago now actively look for inflammatory comments around specific dates: the anniversary of a homicide, say, or the birthday of a slain gang member, the sorts of events that have often incited renewed rounds of violence.”⁵⁹ As an example, he described how a police gang unit notified a 12-year-old’s parents that their son had foolishly insulted a well-known gang member in a posted rap video. Police went so far as to relocate the family temporarily from their home and then watched as angry gang members stalked the young boy’s home. Police believe that this type of preventive, real-time intervention saves lives.⁶⁰

Academics have also begun studying Twitter to predict crime. The idea is that Twitter represents society’s current views, organized by location and content, and thus could be useful for identifying future risks of violence. In an interview with NPR’s *All Things Considered*, Desmond Patton, a professor of social work at Columbia University, explained how

he has developed an algorithm to mine tweets for predicted violence: “One idea is that if we can decode the language, then perhaps we can send triggers to social workers, violence workers who are embedded in these neighborhoods already, so that they can utilize the strategies they already have to reach out to youth before the post becomes an injury or homicide.”⁶¹ While this type of algorithmic mining of content is still in its infancy, one can see that it could offer insights to police about patterns of youth activity in a city. Even something as simple as where the hot parties or hangouts will be on a weekend night might allow police to respond more effectively to prevent out-of-control situations in the immediate future.

But, of course, like in the cellular collection situation, in order to monitor the individualized risks, police and researchers also have to collect everyone’s data. The haystack keeps growing, with billions of bytes of data being shared every day. To find the violent tweets, police also must collect the other content being shared on social media, and this can include First Amendment–protected public speech. In fact, Geofeedia was used to track and arrest young people in Baltimore protesting the death of Freddie Gray.⁶² Similar concerns have arisen after the FBI and the Boston Police Department declared an interest in monitoring public social media posts for investigative purposes.⁶³

Data Mining

“Data mining” involves the search of large data sets to discover new insights.⁶⁴ In policing, this means using data searches to help solve past crimes or uncover ongoing ones.

Data-mining searches have been conducted for years by law enforcement.⁶⁵ Police routinely search for fingerprints or DNA evidence in large national databases,⁶⁶ meaning that pattern-matching software sifts through collected fingerprints or DNA profiles to find a match. The FBI has augmented its search capacity by building a sophisticated biometric database that includes DNA, fingerprints, palm prints, face prints, tattoos, and iris scans.⁶⁷ This Next Generation Identification (NGI) system builds on big data capabilities to allow over 18,000 law enforcement entities to search the growing database.⁶⁸ Similarly, the FBI, the U.S. Treasury, and the Securities and Exchange Commission (SEC) rely on financial

algorithms to track unusual or structured cash deposits, international transfers, or stock manipulation.⁶⁹ These pattern-matching investigative techniques look for suspicious movements in the flow of money and are expanding in sophistication alongside improved technologies.

Data mining also allows for more unstructured searches. For example, when the Richmond Police Department queried its correctional database to study stranger rapes, it found that “a prior property crime was a better predictor of a stranger rape than a prior sex offense.”⁷⁰ As the chief of police explained,

Upon further examination of the property crimes committed by some stranger rapists, we noted that many of their burglaries differed in that they seemed to preferentially target occupied dwellings and frequently stole property with little value, if they stole anything at all. Using this type of anomaly detection, we can identify unusual or suspicious incidents that are worthy of additional investigation and have been able to successfully identify cases associated with an increased risk for escalation based on subtle deviations from normal.⁷¹

This data-driven insight was both counterintuitive and critical to understanding criminal behavior. The result is also something that would have remained hidden in an era without large, searchable data sets.

Data reveals previously unseen connections between places, people, or acts. For example, sex trafficking is a vast and transient business, making it difficult to find the locations of trafficking victims. In response to this challenge, the Polaris Project, a national antitrafficking group, partnered with big data companies to begin geotagging the locations of hotline calls requesting help.⁷² From a data set of 75,000 calls, the Polaris Project began identifying organized networks of trafficking across America.⁷³ Each call could be tagged by location and included the victim’s age, immigration status, language needs, and shelter requirements.⁷⁴ Patterns emerged of certain truck stops that served as the location points for forced trafficking.⁷⁵ Certain cities at certain times (like during Super Bowl week) emerged as places to focus law enforcement attention.⁷⁶ Data mining allowed a geographically fluid and otherwise-secret network to be mapped and financial patterns to be exposed. For example, in targeted areas, nail salons that regularly processed nu-

merous \$100 charges after 11 p.m. were investigated as fronts for illicit prostitution businesses.⁷⁷ Fake entertainment companies used to cover trafficking could be targeted for tax-evasion or money-laundering investigations.⁷⁸ By tracing movement and money through data, a global and quick-moving criminal network could be monitored and prosecuted.

Data mining has also helped police address domestic violence. In New York City, data helped identify those men most likely to escalate violence on their partners.⁷⁹ Over a one-year period, NYPD officers responded to 263,207 calls for reported domestic violence.⁸⁰ Knowing that precursor acts of domestic violence tend to escalate into serious assaults and even homicides but not knowing which of those reports to prioritize, police turned to a computer program to scan the police complaint reports. The algorithm looked for words like “kill,” “alcohol,” or “suicide.” In addition, police studied the addresses of stay-away orders and used those clues to prioritize certain households for additional attention.⁸¹ Building off the data, police created a program whereby designated houses predicted to be the source of future domestic violence were visited and the victims interviewed.⁸² Police conducted tens of thousands of precautionary follow-up visits to deter future acts of violence.

Finally, consumer purchases can be mined for possible clues about illegal activity. For example, one Kentucky study found that sales of over-the-counter cold medicine with the active ingredient pseudoephedrine directly correlated with an increase in meth-lab seizures and drug arrests.⁸³ As reported in the *Los Angeles Times*, “In any given county, an increase in pseudoephedrine sales of 13 grams per 100 people translated to an additional meth lab busted. The results suggest that the computer databases could actually be used to predict where drug busts are most likely to take place.”⁸⁴ More basic correlations can also be made about individuals. By studying the purchase of anyone who bought lye, iodine, ephedrine (Sudafed), Drano, brake fluid, and lighter fluid—all component parts to making crystal meth—police might be able to see future criminal acts from otherwise-lawful purchases.⁸⁵ From the data of ordinary customer receipts, a pattern of criminal activity might emerge.

Of course, many innocent people with colds, injuries, and plumbing or car problems may also buy those fairly basic (and legal) consumer items.

The correlation, while suspicious, is not criminal. And this brings up one of the fundamental questions with algorithmic investigation: can correlation replace causation when the underlying data remains imperfect?

Can Algorithmic Investigation Be Trusted?

The rise of big data policing rests in part on the belief that data-based decisions can be more objective, fair, and accurate than traditional policing. Data is data and thus, the thinking goes, not subject to the same subjective errors as human decision making. But in truth, algorithms encode both error and bias. As David Vladeck, the former director of the Bureau of Consumer Protection at the Federal Trade Commission (who was, thus, in charge of much of the law surrounding big data consumer protection), once warned, “Algorithms may also be imperfect decisional tools. Algorithms themselves are designed by humans, leaving open the possibility that unrecognized human bias may taint the process. And algorithms are no better than the data they process, and we know that much of that data may be unreliable, outdated, or reflect bias.”⁸⁶

Algorithmic technologies that aid law enforcement in targeting crime must compete with a host of very human questions. What data goes into the computer model? After all, the inputs determine the outputs. How much data must go into the model? The choice of sample size can alter the outcome. How do you account for cultural differences? Sometimes algorithms try to smooth out the anomalies in the data— anomalies that can correspond with minority populations. How do you address the complexity in the data or the “noise” that results from imperfect results? The human world remains both complex and imperfect.⁸⁷ The choices made to create an algorithm can radically impact the model’s usefulness or reliability.

To examine the problem of algorithmic design, imagine that police in Cincinnati, Ohio, have a problem with the Bloods gang—a national criminal gang originating out of Los Angeles that signifies membership by wearing the color red. Police use a social media scraping tool like Geofeedia to target possible gang members involved in drive-by shootings. They search key terms like “gang,” “car,” “gun,” “hit,” “cap,” “park,” “run,” “drive,” “shoot,” “strike,” “red,” and “colors.” A series of tweets and social media posts pop up, and police begin tracking links between

the various social media users. On the basis of the identified problem (shootings), the identified targets (people using social media and making gang-related comments), and the location (Cincinnati), the software program comes up with a list of suspects to target. But unfortunately for the accuracy of the computer program, many people in Cincinnati might be talking about the local baseball team—the Reds—using words like “hit,” “run,” “strike,” “cap,” “park,” and other baseball terms to discuss sports. The correlation might be very accurate given the search parameters and algorithmic design but overbroad with regard to finding the actual targets.

Similarly, the data mining of precursor crimes—if designed poorly—can lead to unhelpful and overgeneralized results. PredPol, which has successfully tested place-based predictive policing models of property crimes, attempted to use a similar approach for homicides.⁸⁸ PredPol’s starting point, rather commonsensically, was that if you traced the prevalence of handguns, you could predict future shooting deaths. Find the guns, and you will find the bodies. Using Chicago as its laboratory, PredPol compiled a data set consisting of 38,740 violent crimes during the years 2009, 2010, and 2011. Looking at crimes with “handgun” in the descriptive field, the researchers isolated 17,020 robberies, 6,560 assaults, 8,252 weapons violations, 5,274 batteries, and 303 criminal sexual assaults.⁸⁹ The researchers then compared those precursor handgun crimes to the 1,331 homicides during that same time frame. In a white paper published on the results, the company claimed, “PredPol successfully predicts 50% of gun homicides by flagging in real-time only 10.3% of Chicago. . . . Crimes involving guns continue to have an impact on future gun homicides for 30–100 days and risk spreads over as much as 1/2 mile in area.”⁹⁰ Pulling that conclusion apart, however, demonstrates the misleading utility of the analysis. First, predicting a shooting one month to three months out does not provide much guidance to police on the streets every day. When during that three-month time period should police be vigilant? What would they be looking for during that time? Also the area (10% of Chicago) is a vast area—approximately 23 square miles.⁹¹ Again, without more granular information, this correlation—even if 100% accurate, does not offer enough insight to improve policing.

Algorithms can also just get it wrong. Pedro Domingos, a professor of computer science at the University of Washington and author of *The*

Master Algorithm, recounted an important lesson of mistaken machine learning.⁹² In an article in the *Washington Post*, he explained how a colleague had created a computer algorithm to tell the difference between wolves and wild dogs (which can look quite similar).⁹³ During testing, the algorithm performed perfectly, too perfectly. Only upon further inspection was it revealed why the computer did so well. All of the photos of the wolves had been taken with white snow in the background, and none of the photos of the dogs had snow in the background. The computer had simply learned to identify snow, not wolves.⁹⁴ Similar dangers face predictive algorithms; false flags might prove correlations but only because of a poorly designed model.

Is Big Data Racist?

Sometimes, the machines get it wrong because of racial or gender bias built into the model. For policing, this is a serious concern. As Frank Pasquale has written in his acclaimed book *The Black Box Society*, “Algorithms are not immune from the fundamental problem of discrimination, in which negative and baseless assumptions congeal into prejudice. . . . And they must often use data laced with all-too-human prejudice.”⁹⁵

Solon Barocas and Andrew Selbst are scholars who study the disparate impact of big data technologies.⁹⁶ In an article on why big data discriminates, they articulate the four root problems of all algorithmic decision making: (1) definitional bias, (2) training bias, (3) feature selection bias, and (4) proxy bias.⁹⁷ Combined, these problems contribute to largely unintentional algorithmic discrimination in many big data “supervised learning” models.⁹⁸ Supervised learning algorithms are those that create rank-ordered targets, useful for hiring or matching the probability that someone will commit a crime.

Think about the design problem from a computer scientist’s perspective. Imagine that you are called into an office and told to build a big data system to predict crime. You know how to create a model that predicts outputs from inputs, but there are some big design questions to answer. First, there are definitional questions. Are the inputs crime? Criminals? Patterns? How do you label the crime? What constitutes a burglary (some states include breaking into outbuildings and even cars

as burglaries)? What class of crime does the act fit with? Is a burglary a nonviolent or violent crime? What are the target variables? Do you focus on arrests (which are only allegations of crime) or convictions (which might be resolved only months or even years after the incident)? What types of crime do you care about? The algorithm to predict burglary will look a lot different than the one to predict sexual assault. If you pick a place-based property-crime algorithm to guide your policing strategy, you may be ignoring other crimes in the area that are harder to track. How do you define the location of a crime (block, address, GPS coordinates)? The level of locational precision may impact how useful the resulting prediction model will be for police. These “target variable” and larger “definitional labeling” problems must be answered. Choices must be made. Human decisions impact “target variables” (the answers sought) and “class labels” (the menu of possible answers), and bias from those choices can impact the ultimate algorithmic model.⁹⁹

To build a model, you need to use training data to “train” the model.¹⁰⁰ Algorithms and machine-learning models learn by analyzing past data. This means that the mathematical model’s only ability to learn is to sort through historical data to determine a correlation. Inputs go in, and generalizations come out, so that if historical crime data shows that robberies happen at banks more often than at nursery schools, the algorithm will correlate banks with robberies, without any need to understand that banks hold lots of cash and nursery schools do not. “Why” does not matter to the math. The correlation is the key.

Of course, algorithms can replicate past biases, so that if an algorithm is built around biased data, analysts will get a biased result. For example, if police primarily arrest people of color from minority neighborhoods for marijuana, even though people of all races and all neighborhoods use marijuana at equal rates, the algorithm will correlate race with marijuana use. The algorithm will also correlate marijuana with certain locations. A policing strategy based on such an algorithm will correlate race and drugs, even though the correlation does not accurately reflect the actual underlying criminal activity across society. And even if race were completely stripped out of the model, the correlation with communities of color might still remain because of the location. A proxy for racial bias can be baked into the system, even without any formal focus on race as a variable.

In building a model and training the data, the system requires human judgment. Analysts must make decisions about how to use the data. “Feature selection” has the potential to create bias,¹⁰¹ meaning that the features you choose for the model will alter your result. Again, take the relatively simple model to predict the location of a crime. You can build the crime-prediction model, collect the data, and train the data, but how do you define the resulting output? How big is the predicted high-crime area? Do you choose a 500-by-500-square-foot box, or a half-mile radius? The data could fit both, but the effectiveness and adverse impacts could be quite different for actual policing. How long does the targeted area remain “hot” (a month, a day), and does the strength of the prediction decay over time? Which crimes are weighted as more important to track? These are choices independent to the data but critical to the usefulness of the model.

Finally, proxies for racial bias can infect the data. Barocas and Selbst discuss the problem of “redundant encodings,” when “membership in a protected class happens to be encoded in other data.”¹⁰² It may be that the data has been trained to look only at neighborhoods, but if city neighborhoods have been segregated through racially discriminatory zoning laws or have been economically fractured such that poverty, race, and place correlate, then a place-based predictive model may unintentionally discriminate. Data-based systems import the biases of their builders and the larger society. Data is not blind. Data is us, just reduced to binary code.

These concerns with predictive big data technologies appear in most big data policing models. Worse, the bias in the system can justify unequal or unfair treatment by police. If the algorithm is guiding policing, then police administrators and patrol officers are likely to follow. Thus, as Jules Polonetsky, Omer Tene, and Joseph Jerome have recognized, “big data could have the perverse result of *exacerbating* existing inequalities by suggesting that historically disadvantaged groups actually *deserve* less favorable treatment.”¹⁰³ If one does not see the potential bias in the inputs, all that is left is a seemingly neutral objective output.

As algorithms become more central to our lives, the racial biases built into them have been exposed. Notorious examples involve an experiment by Harvard professor Latanya Sweeney, who demonstrated that Google’s AdWords sales program discriminated by race.¹⁰⁴ Sweeney

studied 120,000 internet-search ads promoted by data services that provided criminal background checks. When names associated with someone of African American descent were queried, the ads displayed for criminal background checks. When names associated with the white community were queried, no such ads displayed.¹⁰⁵ Another Google algorithm has been called out as sexist. Researchers at Carnegie Mellon created a tool that showed that Google search queries promoted prestigious job openings to men more than to women.¹⁰⁶ This was not because Google programmed its search algorithm to be intentionally sexist, but the underlying data reflects existing gender bias in society. Simply taking gender or race out of the model does not cure the underlying bias. As mathematician Jeremy Kun has written, “It’s true that an algorithm itself is quantitative—it boils down to a sequence of arithmetic steps for solving a problem. The danger is that these algorithms, which are trained on data produced by people, may reflect the biases in that data, perpetuating structural racism and negative biases about minority groups.”¹⁰⁷ Big data policing involves a similar danger of perpetrating structural racism and negative biases about minority groups. “How” we target impacts “whom” we target, and underlying existing racial biases means that data-driven policing may well reflect those biases.

Probabilistic Suspicion?

Data mining cell-tower numbers or metadata or crime data all rely on the same principle: there is a high probability that suspicious linkages will connect a suspect to a crime. The probability that the High Country Bandit’s cellphone coincidentally would be at each of the banks at the exact time of the robberies is just too high to ignore. The probabilities suggest criminality and to a high degree of certainty.

But should an algorithm be sufficient to generate probable cause for an arrest? Should an algorithm that alerted police to your purchase of fertilizer, propane tanks, and matches, plus a rental truck, be sufficient to generate an arrest warrant on the suspicion of creating a truck bomb like that used for the Oklahoma City bombing of the Alfred P. Murrah Federal Building? While everyone might want the danger investigated, does it rise to the level that you can be actually handcuffed and brought to jail? Or stopped on the highway? What if the algorithm said it was a

90% certainty? Can any well-designed, well-trained, bias-free algorithm with high levels of accuracy be enough to generate an arrest warrant to result in a physical, real-world detention? This is the ultimate data-driven policing puzzle.

Two contradictory intuitions collide when thinking about pure probabilistic suspicion. The first involves acknowledging that prediction has long been a part of law enforcement, and predictions, almost by definition, can be wrong.¹⁰⁸ The second involves the discomfort felt about generalized suspicion. When it comes to liberty, mere correlation or a mathematical probability should not, the thinking goes, be enough to justify police action. Both sides can claim some support from the case law, and neither side has won the day in court.

First, unquestionably, the Fourth Amendment recognizes the need for predictive suspicion. When police officers draft a search warrant, they are making a prediction that contraband will be located at that particular place.¹⁰⁹ The judge must determine whether “there is a fair probability that contraband or evidence of a crime will be found in a particular place.”¹¹⁰ It is always possible that the contraband will be gone or was never there in the first place. Predictive suspicion is, thus, not a new concept, tracing its roots back to the words “probable cause” in the text of the Fourth Amendment. As the Supreme Court stated, “In the typical case where the police seek permission to search a house for an item they believe is already located there, the magistrate’s determination that there is probable cause for the search amounts to a prediction that the item will still be there when the warrant is executed.”¹¹¹

Probable cause, as the term suggests, turns on probabilities. And these predictive probabilities are not necessarily the stuff of fixed numbers. In fact, the Supreme Court has refused to provide clear percentages as to the certainty required for probable cause.¹¹² The Court has stated, “The probable-cause standard is incapable of precise definition or quantification into percentages because it deals with probabilities and depends on the totality of the circumstances.”¹¹³ The reason for this lack of guidance is largely because judges must make quick decisions with imperfect information, and holding courts or police officers to a particular percentage seems both unfair and unrealistic.¹¹⁴

This deference continues on the streets, where investigative detentions (police stops) also involve quick predictive judgments. The stan-

dard from *Terry v. Ohio* states that to justify a stop, police must “be able to point to specific and articulable facts which, taken together with rational inferences from those facts, reasonably warrant that intrusion.”¹¹⁵ In other words, police must articulate why they predict a certain person is involved in criminal activity. Again, the prediction can be wrong, but at its core, it is a predictive, probabilistic judgment.

Predictions also underlie forensic investigation. A DNA match is, in reality, a probability that the two biological samples match.¹¹⁶ The prediction can be quite strong (with a low chance of error), but it is still a prediction. Strong predictions can even be used to justify an arrest warrant—for example, in a “cold case” where a DNA match is the only thing linking the suspect to the crime scene. Police reliance on predictive theories and technologies are thus neither new nor necessarily controversial.

Background principles about prediction, however, run into our discomfort of generalized suspicion. In each of the examples just given—arrest warrant, *Terry* stop, DNA match—the prediction is linked to an individual person suspected of a crime. There is a reason why this person or that house or that biological material should be searched. But what about a situation in which all you have is a generalized suspicion, albeit with a high probability of accuracy? In an article on the Fourth Amendment, Arnold Loewy offered an intriguing hypothetical question on this point: “Suppose that in a particular city block of Main Street, between Fourth and Main and Fifth and Main, it could be established demographically that nine out of every ten men on the street between 6 p.m. and 10 p.m. are carrying drugs. Would that create probable cause (or reasonable suspicion) to arrest any man found on that block of Main Street at the requisite hours?”¹¹⁷ By any measure, a 90% probability would easily surmount the probable cause standard required by the courts. Yet something feels wrong. The generality of the suspicion, which very likely will target innocent people (one in ten are not guilty), cuts against the support of otherwise-strong probability-based suspicion.

What is wrong is the lack of individualized suspicion. As Loewy answered his own question, “The answer, I believe, is ‘no.’ Probable cause and reasonable suspicion require more than demographic probabilities. There must be something specific to the defendant to create

the probability as to him (perhaps a furtive gesture, an informant's tip, excessive nervousness, etc.)."¹¹⁸ Something more than generalizations about the "high crime" location or the "hot" people who frequent the area must be established. As the Supreme Court said in another case, "probable cause requires 'that the belief of guilt must be particularized with respect to the person to be searched or seized.'"¹¹⁹ Or as Tracey Maclin put it in historical context, "It is a fair summary of the history of the Fourth Amendment to say that the provision reflected the Framers' desire to control the discretion of ordinary law enforcement officers and to eliminate governmental intrusions lacking particularized suspicion."¹²⁰

Applying this principle of particularized suspicion to the problem of pure probabilistic data mining, courts may be reluctant to allow generalized suspicion (even with a high degree of accuracy) to constitute probable cause. Something more, individualizing the suspicion to a person, may be necessary to satisfy the Fourth Amendment. So database suspicion alone should be insufficient to meet the constitutional threshold.

But this requirement presents difficult questions. For example, imagine that an algorithm alerts when any white man between 40 and 60 years old files bankruptcy, receives emergency psychiatric services, and buys a gun in the same week, on the theory that in combination, those life stresses and actions are statistically more likely to result in impulsive, emotional acts of violence. Assuming that there is clear data that such clusters of events result in violence (as of now, there is no such data), should police act to stop, seize, or even arrest the suspect? One difficulty is that the judge or arresting officer will have no way to test the accuracy of such a correlation. The numbers remain a black box. Another is that the generic suspicion that may attach to such a cluster may not attach to this particular suspect. An innocent man who is facing already-significant life issues will now face criminal suspicion. A third issue is that no one in the system knows what to do with the information. One solution, of course, would be to have the alert lead only to a contact, not any actual criminal action. Police might call or inquire but not do anything more invasive. In this way, the algorithmic alert would serve more of a public health function, rather than a law enforcement function. But, again, one can only imagine the response of this already-stressed individual when the police knock on the door to "talk" about his problems.

Other scholars have pushed back against the whole conception of individualized suspicion.¹²¹ After all, if you consider how police really make decisions, they routinely generalize and use intuitions from past experiences to predict future actions. Individual judgments, in truth, are just reflections of generalizations from past experience.¹²² For example, our understanding of concepts like “pretty” or “ugly,” “suspicious” or “innocent” only come from our own personal algorithm of accumulated past experiences and intuitions. Bernard Harcourt and Tracey Meares detail a more accurate reality of police suspicion:

Suspicion attaches to a group trait that an individual displays, such as having a bulge in one’s pants pocket, fitting a description in the vicinity of a recently committed offense, throwing away a plastic vial at the sight of a police patrol car. . . . These are group-based determinations often made irrespective of the officer’s knowledge of whether a specific offense has been committed, and suspicion potentially attaches to all individuals within these categories. Suspicion in these cases is “individualized” only in the sense that it attaches to an individual because he or she is a member of the suspect group. In other words, in most cases of policing, suspicion does not originate at the individual level.¹²³

In other words, individual suspicion really incorporates generalizations about group action. Police use these cognitive shortcuts all the time, and if police can use these group traits for suspicion, why should algorithms not also be able to draw similar group correlations together? The difficulty, as has been discussed, is that these shortcuts or generalizations include implicit (and sometimes explicit) bias, and the same problems that infect human generalizations can also infect algorithmic generalizations. Just because police generalize does not make it permissible to design an algorithmic system to mirror the same biases.

The legal question about pure algorithmic suspicion remains open. Daniel Steinbock has argued, “Predictive profiling is not inconsistent with the Fourth Amendment, [and] the factors used must indicate to the investigating officers (and, later, the reviewing court) the requisite degree of suspicion. Nothing suggests that these actors should defer to a computer algorithm for projecting that level of suspicion, but nothing rules out that possibility either.”¹²⁴ Perhaps improved big data technolo-

gies will allow us to feel more comfortable with this type of generalized but accurate correlation. I have my doubts, as the human requirement to confirm suspicion seems necessary in a human system of law enforcement. Even if the algorithm or alert system can help find the needle in the haystack, a human should confirm that the correlation makes sense in this particular case. But courts have yet to answer this question, and the existing law may well allow it. This will matter because the rise of data mining and the ease of matching those data-driven clues to real people will only increase in the future.

How Big Data Impacts How We Police

Data collection, data mining, and data analysis shape both police investigation and police practice. The work involves technical expertise and a significant investment in technology.

For investigation purposes, the move toward data mining opens up a new type of suspicion. Data creates the clues. Data generates the leads. Human suspects emerge only after the data has been queried. As Erin Murphy has recognized, “The use of databases to generate suspects represents a new kind of investigation altogether—whether based on particular information (e.g., “who called this number”) or upon predefined algorithms (e.g., “who has traveled to these three countries and bought these two items within a one month period”).”¹²⁵ This type of suspicion offers the ability to develop targets without the report of a crime. Building off the proactive, intelligence-led investigative model, this type of predictive suspicion will result in more surveillance and a greater need for police to collect data. If you combine the various different surveillance technologies discussed in this book, you can see that a huge change awaits law enforcement. Although analyzed separately, each of these technologies—person based, place based, real time, social networks, aerial surveillance, and so on—can be aggregated in ways such that data mining will offer new possibilities to develop suspicion.

More practically, police administrators will need to begin hiring more technologically savvy data geeks. Unless police agencies wish to outsource their investigation to private companies and consultants to manage the technology, then in-house officers will have to be trained (and retained). Currently, the public-private partnerships between LAPD and

Palantir or NYPD and Microsoft (which built the Domain Awareness System) appear to be working, but one can imagine all sorts of problems in the future. The private company could decide not to continue working in the policing space, abandoning the technology and the city contract. The private company could itself be suspected of criminal activity (creating a major conflict of interest). Budget issues could require police to scale back the participation arrangement, and without continually trained, data-smart officers, a data-dependent policing system could fail.

Changing how we police also involves a change in mind-set. Police administrators face a big challenge to get officers to buy into big data's potential. Policing is a status quo job, which by design and tradition protects the status quo. Many police officers have practiced within traditional policing structures for decades, so moving to a data-driven system brings resistance, distrust, and tension.¹²⁶ Early versions of predictive policing technologies have been met with skeptical tolerance, with senior officers thinking the technology adds little to their own professional knowledge of the area and others thinking that the products are more magic than science. Police administrators thus must educate officers of the value added by data-mining technology.

Finally, data-driven policing will require a focus on precision. Police administrators will need to develop quality-control mechanisms for data-mining systems. Big data in the consumer space does not need to be perfectly accurate to be helpful, but big data in the policing space does not have the same luxury to tolerate error. As big data systems grow in scope, more errors will be included, and government officials must be able to cleanse, cure, and trust the data. It cannot be the case that a chief of police has no idea of the content or reliability of the information that is influencing his or her officers. It should not be the case that the police cannot reveal the nature of the technological tools used to capture information. As attractive as new surveillance tools may be, police must take ownership of their use and be able to defend their legality. Police departments cannot live in the dark.

Black Data

Distortions of Race, Transparency, and Law

Seeing in the Dark

In the near future, the innovations of big data policing will amplify “who,” “where,” “when,” and “how” law enforcement targets crime. While not all jurisdictions will utilize all of these new technologies and while new technologies will continue to develop, every police force using big data techniques will share one thing: a “black data” problem.

“Black data” denotes three overlapping concerns with the rise of data-driven policing, involving race, transparency, and constitutional law. First, data comes from the real world, weighted with racial disparities and a history of unfair application. Second, new data technologies are secretive, scary, and unfamiliar. There exists a darkness. A void. A fear. We cannot see into the future. And, third, what we do see regarding privacy or constitutional protections seems distorted.

The tensions inherent in black data will shape perceptions about new policing technologies. At this moment, at the intersection of new technology and old policing problems, black data must be illuminated. This chapter seeks to examine the future of big data policing through the lens of black data. Only by confronting the issues of race, transparency, and constitutional distortion can a future vision of big data policing become clear. Like one’s eyes adjusting to a dark room, seeing in the darkness of black data allows a differentiation of those shapes that one should be scared of and those that are harmless or even helpful.

Black Data and Race

Big data’s claim to objectivity and fairness must confront the racial history of American policing. Police data remains colored by explicit and

implicit bias. Police data is racially coded, shaded by millions of distrustful looks and thousands of discomfiting physical encounters. The data incorporates the lived experience of African Americans, Latinos, and other people of color who have rightly or wrongly felt discriminated against because of society's suspicion.¹ In short, big data policing must acknowledge that race is still part of modern policing. Even as code, the data is black.

Acknowledging the “blackness” of police data serves two very useful goals. First, this admission pushes back against the facile claim that the issue of race can be removed by “objective,” data-driven policing or that new big data technologies avoid racially biased effects. Second, the acknowledgment forces a serious examination of the very human injustices that still stir resentment and distrust in minority communities. Police can develop technologies to address and overcome these racial tensions, but first the adopters of data-driven systems must acknowledge and design around this racial reality.

As detailed throughout the book, big data policing reifies many of the systemic inequalities of traditional policing. Place-based predictive systems culled from unrefined crime statistics reflect policing patterns, not necessarily the underlying crime patterns.² The choice to send undercover narcotics investigators to inner cities as opposed to Ivy League colleges ensures drug arrests of the poor instead of the privileged. Targeting high-risk blocks because of the number of returning citizens or people on probation living there encodes the prior decisions of a criminal justice system that disproportionately punishes people of color. Suspect-based predictive policing, because it takes into account prior arrests, suspected gang associations, and the like, cannot help but also reflect the socioeconomic impacts of traditional policing patterns.³ As none other than Jeff Brantingham, the father of modern predictive policing, warned about person-based models, “These ‘person-centric’ models are problematic . . . because they carry an elevated margin of error and can legitimize racial, gender-based and socioeconomic-driven profiling. As a scientist you better be damn sure the model of causality is right or else it’s going to lead to a lot of false positives.”⁴

Other systemic racial inequalities exist. One can see how the pressure to collect “field interview cards” can create a self-fulfilling system of suspicion for those high-risk individuals targeted.⁵ Official policies

of aggressive stops in minority neighborhoods generate future suspicion shaped in part by the number of prior contacts. Even pure algorithmic suspicion, which should avoid charges of racial bias, can be found to be infected with biases due to the way the models are built or the data collected and trained. If facial recognition, pretrial risk assessments, and Google search query technology cannot avoid race-based disparate effects, then all data-driven technologies must be alert for potential bias.

So, while promising a clean start, big data policing cannot—in truth—escape a long history of racial discrimination. Instead, it must confront an uncomfortable tradition of social control in minority communities. As Sandra Bass has summarized,

The interactive relationship between race, space and policing has been of social and political significance since the earliest days of American history. Monitoring the movement of slaves was a central concern for plantation masters and slave patrollers. The desire to regulate and subjugate the behavior of newly manumitted slaves was the primary impetus for creating new legal rules against vagrancy and loitering in the post-antebellum South. The rise of Jim Crow and the location and construction of urban ghettos and public housing were deliberate efforts to promote social control and isolation through racial containment. For the better part of our history, race has been a central determinant in the definition, construction and regulation of public spaces.⁶

For over a century after the Civil War, local police enforced racially discriminatory laws.⁷ Police captured escaped slaves. Police enforced economic peonage. Police failed to protect civil rights protesters.⁸ Police allowed violence, lynchings, and prejudice to undermine the criminal justice system.⁹ It is no accident that the major urban riots from the 1960s to the present day all arose in anger to instances of police brutality.¹⁰ The petty harassment of Ferguson, Missouri, or the repetitive disrespect of NYPD's stop-and-frisk program or the death of unarmed men in police custody all reclaim—in the present day—a shared history of racial discrimination at the hands of the police.

And these dark marks exist branded with official sanction. The racial harm was inflicted knowingly. In *Terry v. Ohio*, the case that con-

stitutionalized stop-and-frisks in 1968, the Supreme Court conceded the future racial impact on overpoliced minority communities: “The frequency with which ‘frisking’ forms a part of field interrogation practice . . . cannot help but be a severely exacerbating factor in police-community tensions. This is particularly true in situations where the ‘stop and frisk’ of youths or minority group members is ‘motivated by the officers’ perceived need to maintain the power image of the beat officer.’”¹¹ The NYPD stop-and-frisk program, declared unconstitutional in 2013, systematized a modern version of this “power image of the beat officer.” The constitutional wrong involved a system of control using the legal authority and physical force of police against particular communities.

This “black” reality of social control literally and figuratively colors the introduction of big data technologies in the modern era. The history of American policing includes the history of surveillance of African American communities and civil rights protests.¹² The FBI in the 1960s specifically targeted civil rights protesters for government surveillance¹³ as agents of the state imprisoned Martin Luther King Jr. and many others in order to silence a message of racial equality. Phones were bugged, lives examined, meetings infiltrated. From W. E. B. Du Bois to Fannie Lou Hamer to the Movement for Black Lives, police have sought to monitor threats to social order through surveillance.¹⁴ So why should big data technologies be different when they emerge from such a dark history? How can big data surveillance avoid the traps of the past, when initial uses—“predictive heat lists” filled with nothing but poor people of color, African American activists monitored on social media, sentencing risk assessments found to be racially discriminatory, and surveillance flights only over West Baltimore—all point to a high-tech form of parallel discrimination?

Answering these questions will shape the future of big data policing. Fortunately, the questions are starting to be addressed. In 2016, Paul Butler and Alvaro Bedoya hosted a conference, “The Color of Surveillance,” at Georgetown University Law Center that focused on the intersection of surveillance and race. Scholars, activists, technologists, the general counsel of the FBI, and a national audience began the process of acknowledging past police practices with an eye toward the technological future. Bedoya explained the purpose of the conference:

We're having the most intense surveillance debate in a generation. But that debate largely misses the point that when everyone is watched, not everyone is watched equally. The burdens of surveillance have historically and to this day been disproportionately borne by African Americans and other "others" that society deems dangerous. The justification for that surveillance has been both law enforcement and national security. If we want to create meaningful protections for privacy and civil liberties, we have to reckon with the color of surveillance.¹⁵

These types of events, engagements, and forums will ensure that big data policing does not remain anchored to the past. By acknowledging the impact of race on black data policing, a space can open for discussion. Or as Butler reflected, "The race crisis in criminal justice is not just about police and incarceration, it's also about surveillance in its more technological forms. The voices, history, and moral authority of African Americans are crucial to the discourse on privacy and technology."¹⁶

Acknowledging race in policing data requires new technological capabilities. Both the inputs and outputs of big data policing must be monitored for a disparate impact on communities of color. Clearly, explicit uses of race can be kept out of computer models, but race is too easy to mask as something else. Choices must be interrogated to ensure that race does not impact police decision making, which is easier said than done. Arrest statistics might be simple to collect and measure, but a system that looks only at arrest statistics applied to a city like Ferguson, Missouri, will lead to racial inequity.¹⁷ Such a system would measure arrests at the expense of actual crime drivers. Such a system would confuse a desire for municipal revenue (in the form of arrests/fines) with actual crime, thus disproportionately impacting people of color.¹⁸ Changing the inputs from arrests to reported calls for service (reported crimes) or removing low-level arrests from the inputs could alter the outcomes.

Designing data systems to avoid racial bias has begun in other professional fields.¹⁹ Private companies now use algorithms for hiring decisions, because sorting through a stack of applicants can be done relatively easily by a computer model that examines past successful hires and then looks to replicate those hires from résumés submitted. Of course, bias can creep in; if most of the company's past hires were

white, Ivy League-educated men, then training the data to match past hires could result in a racially imbalanced hiring strategy. Recognizing this reality, scholars like Ifeoma Ajunwa have proposed a system to examine hiring algorithms and evaluate them for racial bias.²⁰ Using such a technique, companies can be comforted that the hiring algorithm did not unintentionally discriminate based on race or gender, and if bias is identified, the model can be modified. Such a process, because it focuses on the model itself, as opposed to the inputs, might also be applicable to policing models. Cities could request that companies certify that their predictive policing systems avoid disparate impacts.²¹

The blackness of big data policing cannot be easily fixed. But one important lesson can change how big data policing moves forward in the future. The lesson is simple: remember race. Despite the temptations to focus on the math or the technology or the futuristic sound bite, big data policing will remain black (and brown). Being conscious of potential racial bias does not delegitimize big data policing but merely offers a constant corrective reminder that the implicit and explicit biases that impact policing on the streets also impact the black data in the system.

Black Data and Transparency

Beyond race, big data policing also suffers from a transparency problem. With the exception of data analysts, almost all data systems are black-box mysteries to the user.²² Proprietary predictive algorithms protect their secrets.²³ Open-source predictive algorithms remain inscrutably complex. Even simple databases cannot be easily understood by the end user. Policing is not a field that typically requires PhD-level technological expertise, and while some data scientists participate in developing police strategy, the vast majority of police administrators and patrol officers do not have the code-writing capacity to look under the big data hood. In addition, the black data being used by police to monitor criminal activity needs to remain secret enough to be effective. For both officer safety and tactical advantage, data secrecy matters.

The concept of black data symbolizes this problem of transparency and lack of accountability. If police, courts, or citizens cannot understand the technology and if lawyers, journalists, and academics cannot interrogate the data, then how can anyone trust it?

Big data policing must address this opacity. Secrets live in the dark. Secrecy distances citizens from publicly accountable government entities and heightens confusion about the technology. Black data policing, thus, creates a legitimate fear that secret systems will impact people's lives and liberty. Citizens rightly may question why the tools used to protect them cannot be explained to them. Communities may fairly express skepticism that secret technology that cannot even be revealed in court (like the stingray device) works to their benefit.

Police seeking to overcome the black data transparency problem must first orient citizens to the fact that big data policing is not too different from other algorithmic decisions in their lives. Algorithms are already controlling what we read on our Facebook feed, what our credit looks like at the bank, what opportunities exist in the job market, what medicines are prescribed, who gets audited by the IRS, which insurance rates cover us, what consumer coupons we receive, and even the answer you get when you query Google for "what do algorithms control in daily life."²⁴ Admittedly, with the exception of the taxman, none of those other decisions inflict as much physical or psychological discomfort as may arise after a visit from the police, but acknowledging this reality may allow people to come to terms with the fact that much of our lives are controlled by nontransparent algorithms and scoring systems.²⁵

Second, police must see that the response to a lack of transparency is not more information but more accountability. Citizens do not necessarily need information about the mathematical formula underlying the algorithm but do need explanation about why the algorithm is being used and what mechanisms exist to hold the creators accountable. Revelations about secret stingray devices or the secret NSA metadata program generated far more controversy than the equally privacy-invasive Domain Awareness System in New York. The outrage came from the secret nature of government surveillance, more than the actual technological surveillance capacities.

The public health model of identifying environmental health risks provides a helpful analogy. Imagine that a city realizes that its public water supply is potentially contaminated with lead pipes installed in poor neighborhoods. The pipes are hidden, irregularly placed, and limited to only some homes. The city knows the likely risk of homes by reviewing old work contracts, shipping supply receipts, out-of-date plumbing

maps, and a host of other imprecise variables. The identification process is a guessing game ordered by a simple algorithm to figure out the risk of homes and then separately the risk of children of a certain age in those homes. Creating a secret, largely incomprehensible algorithm to rank children most at risk of developing lead poisoning would be seen as unobjectionable even though personal data is exposed, children are targeted, the math is complex, the formula is hidden, and the outcome is perhaps racially skewed to impact poor children of color. Yet a similar algorithm to identify young men who might be victims of violence triggers fear and questions about transparency. The issue—in reality—is not the transparency of the algorithm (in both cases, we might not understand the math) but the transparency of how the program is explained to the public and, of course, what is done with the information.

An emphasis on algorithmic transparency, thus, may be misplaced for two separate reasons. The first is political, the second technological. Politically, as seen in the public health example, what matters is the reasons given to the community for why the algorithm is used. The issue is explaining not the mathematical formula but the reasons why a mathematical risk assessment is necessary (to find those who are most at risk of lead poisoning). Transparency in explanation for why the nontransparent algorithm is being used may be more valuable than the actual schematics for the computer model. Sitting down with a mother and explaining why her son's three prior arrests, friendship with a homicide victim, gang ties, and truancy from school make him a target of violence provides the transparency needed. His raw "heat-list score" provides very little additional clarity and the underlying rank-ordering formula even less. In Chicago, the plan for individualized meetings, custom notification letters, and personalized threat assessments recognized the value of procedural fairness and communicating the reasons behind the targeting system to the targets.

The second reason why demanding algorithmic transparency may be misguided is that as a technological matter, it may be impossible. Many of the predictive policing technologies are owned by private companies whose business models depend on keeping proprietary technology secret.²⁶ Revealing the source code means revealing the company's competitive advantage in business. Other algorithms are built using machine-learning tools that by design learn from prior modeling and,

thus, never repeat the same calculations.²⁷ In artificial intelligence, machine-learning models, the analysis will be different each time because the machine will have learned from the last time it processed the information.²⁸ Doing calculations millions and millions of times means that the last calculation is different from the one before. Such a constantly evolving technology means that one cannot—even with the technical abilities—see the underlying formula because the system is not static. Nor can you audit the artificially intelligent model in the traditional way because the model will always be a moving target.²⁹

Calls for transparency, thus, may not be possible in the sense of revealing the inner workings of the machines. But such calls also may not be necessary. What matters is not the math but the design and proving that the design of the model is fair, is accurate, and works. A group of computer scientists and scholars—Joshua Kroll, Joanna Huey, Solon Barocas, Edward Felten, Joel Reidenberg, David Robinson, and Harlan Yu—collaborated on an article that addresses this challenge of looking for transparency in an ever-evolving black-box system.³⁰ They propose a solution that involves designing computer systems with “procedural regularity,” which means revealing how a computer system makes decisions and then creating mechanisms to test whether the computer system worked as designed.³¹ Using tools of computer science—complicated things like “cryptographic commitments,” “zero-knowledge proofs,” and “fair random choices”—a predictive model can be designed to be tested for fairness without having to give up the source code.³² For our purposes, what needs to be understood is that computer scientists can test whether a model works, even if they do not know exactly how the model works. By designing a model that can be tested, the most advanced machine-learning systems can be held accountable, even though they remain decidedly not transparent.

In the end, all of the fancy math comes down to the question of accountability. Does the model work, and how can we be convinced it does so in a fair manner? Accountability rather than transparency provides a way out of the black data opacity problem. Accountability by design will require big data policing to confront the problems raised in this book about bias, fear, and fair application. Some of this accountability may require additional communication—for example, explaining the inputs, explaining the design, and explaining how the system was designed to

avoid racial or other bias—but this does not require complete transparency. True, certain predictive models may fail this basic requirement. Those companies or programs that refuse to disclose inputs, theories, or how they avoid bias may not be able to overcome the black data problem. But others may rise and succeed precisely because they embrace the challenge to explain how their predictive models work and why they should be trusted.

Black Data and Law

The law must also confront the distortions of big data policing. Data-driven changes of police role, practice, and culture have blurred traditional legal analysis and weakened existing constitutional protections.

Police action ends up in court. Policing technology gets litigated in court. Judges, prosecutors, and defense lawyers will be confronted with the problem of black data in the near future, and their jobs will be made more difficult because big data surveillance threatens to significantly reshape the existing Fourth Amendment doctrine.

As has been discussed, new surveillance technologies alter understandings of a reasonable expectation of privacy. With a Domain Awareness System recording your movements 24/7 or ALPR tagging your car everywhere you drive or facial-recognition technologies marking your location, how can one claim any expectation of privacy in public?³³ Five justices of the U.S. Supreme Court expressed a willingness to protect long-term (28-day), aggregated surveillance through GPS tracking, but real questions remain about short-term surveillance.³⁴ We simply have no clear answer to whether things like pervasive, high-altitude video surveillance violates the Fourth Amendment. Nor has Congress stepped in to provide statutory clarity.

Big data also skews the reasonable-suspicion decision to stop individuals on the street. The algorithmic decision to place someone on a Chicago-style heat list could mark that person for less Fourth Amendment protection.³⁵ After all, how does a judge deny the reasonable and articulable suspicion of an officer who sees the “number four”-ranked person in all of Chicago and stops that person on suspicion of being involved in violence? As a constitutional matter, the ranking should not

be enough to justify the stop, but as a practical matter (in the crush of criminal court), it will likely influence the judge to find suspicion. With more information about individuals, the rather-weak limitations of reasonable suspicion will fall away. Innocent factors cobbled together (friendships, neighborhood, clothing, social media connections) will be crafted into suspicion as necessary. The result of a “small data” doctrine confronting a big data world will be less constitutional protection.

Similarly, the predictive policing technologies that can mark out areas of suspected crime may also mark out areas of lesser Fourth Amendment protection.³⁶ The “high crime” designation may merge with the predictive forecast, and suddenly an algorithm will alter the constitutional protections in a neighborhood.³⁷ Again, what is a judge expected to do when she learns that the stop occurred in a predicted “red box” of crime and the officer based his suspicion in part on that prediction? Factoring in that information to the totality of circumstances will add weight to the officer’s otherwise-unconstitutional hunch. Predictive technology will impact the Fourth Amendment protection for all persons located in that targeted area.

Finally, questions of probabilistic suspicion will bedevil courts. Precise algorithms providing accurate but generalized suspicion will make decisions on probable cause very difficult. Metadata will provide actionable clues that police will want to act on in real time. Courts will have a difficult time saying highly predictive, pure-probability suspicion is not good enough. Unmediated social media threats or admissions of guilt will raise new issues of whether personal online communications are enough to support probable cause. Warrants for cell towers, stingray intercepts, and biometric samples will become more and more routine. Each of these technologies will face a certain legal challenge and an uncertain legal future.

Fortunately, the constitutional problem is probably the easiest of the black data problems to solve. Courts have regularly faced new technologies. Cars, telephones, and smartphones were all once “new.”³⁸ While slow, judges have largely managed to adapt old law to new technology. In fact, courts are pretty well suited to the slow evolution of legal principles and are certainly well practiced. Legislative responses on occasion have also filled in the legal gaps, and law professors like myself have made careers theorizing how the Fourth Amendment fits these new technolo-

gies. While it is true that the fear of the new is disorienting, as in the past, the Fourth Amendment will adapt and change and respond to the challenge.

Overcoming Black Data

Big data policing is here to stay. Beyond being helpful, the technology is simply too innovative and futuristic to ignore. The appeal of new technological solutions goes deep in our psyche. Americans in character and temperament remain fascinated by “the next new thing.” Splashed across every magazine, newspaper, and television news show is a headline about the “next breakthrough in science or technology.” A country of upstart rebels, restless innovators, and progressive thinkers has always had one eye on the future, which is why police, like the rest of us, have embraced the idea of big data so effortlessly. The insights and advantages of new data-driven policing remain just too tempting to ignore.

To get to that promising future, police must overcome the black data problem. Black data policing must absorb the complexity of new technologies and old practices. Police must respond by acknowledging challenges and addressing the complexities of racial bias, a lack of transparency, and legal distortions—good and bad. The black data problem can be overcome but only by confronting the complexities rather than blindly pretending to turn the page to something new. In this way, a light can shine on the blackness and reveal a deep and much-less-frightening future.

Big data’s potential, however, goes further. The next chapter examines how big data could revolutionize policing practice, by inverting the architecture of surveillance to collect blue data systems of police misconduct and racial bias.